

Electroweak Mass Ratios from an Internal $SU(2)$: A Kaluza–Klein Reading of the de Vries Relation

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(preprint; the status of each claim — theorem, derived, structural, or conjectural — is stated in the text)

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Abstract

The on-shell electroweak mixing angle, $\sin^2 \theta_W = 0.2231$, and the mass ratio M_W^2/M_Z^2 are reproduced to a few parts in 10^4 by a single algebraic object with no continuous free parameter: the positive eigenvalue of the 2×2 “branch operator” $\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}$, evaluated at the Casimir labels $J = j(j+1)$ of the spins $j = 0, \frac{1}{2}, 1$. This is the empirical regularity — the *de Vries relation* — that we take as the clue fixing the internal geometry. The central claim of the paper is that the internal manifold we live on is essentially fixed by the Standard Model gauge content: the ten-dimensional Type IIA geometry $M_4 \times K_6$ with $K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1$ — the minimal product of the minimal $SU(3)$ coset (colour, four-dimensional) and the minimal $SU(2)$ coset (weak isospin, two-dimensional), saturating the internal dimension $4 + 2 = 6$. (We are careful with “unique”: this is the minimal product realization, not a proven-unique manifold; see Section 7.) We show that $\mathcal{A}(J)$ is the linearized flux-fluctuation spectrum on this K_6 , and that its single scale modulus is what grows a seventh dimension (recovering the well-understood eleven-dimensional M^{pqr} of the 1980s, here a mere limit) and shrinks a fifth (the nine-dimensional infrared endpoint). Three statements organize the operator. First, a short representation-theoretic argument shows that the labels $j = 0, \frac{1}{2}, 1$ are exactly the half-integer fingerprint of an $SU(2)$ (rather than $SO(3)$) internal geometry, and that $J = j(j+1)$ is one quarter of the scalar Laplacian eigenvalue on the unit S^3 . Second, the off-diagonal $\sqrt{J}$ is the matrix element of a *first-order*, self-adjoint, $SU(2)$ -covariant operator — concretely the curl $\star d$ on co-exact one-forms, whose square is the Hodge Laplacian; this is why the mixing is a square root of the Casimir and not the Casimir itself. Third, the negative diagonal is the form-degree mixing induced by the ten-dimensional Type IIA background flux on K_6 , with a coefficient c whose self-dual value $c = 1$ gives the de Vries block. We prove a normal-form theorem: a minimal set of axioms (gauge masslessness, Casimir factorization of the mixing, a single generating operator, no free dimensionless constant) forces $\mathcal{A}(J)$ uniquely. We are careful not to overstate what this buys: the weak angle is determined with no *continuous* free parameter, but only *conditional* on the discrete inputs (the electroweak spin content $j = \frac{1}{2}, 1$ and the coefficient $c = 1$). For $c = 1$ itself we give a *conditional derivation*: we prove that $c = 1$ is forced if the order parameter is a flat geometric modulus coupling to the flux through a single Stückelberg portal (so the order-parameter mass is $-A_j A_j^\dagger$, built from the same operator as the mixing), and that this $c = 1$ is robust against the Weitzenböck curvature shift; we also show that the alternative — an independent, curvature-shifted order-parameter mass — gives a j -dependent $c = (4J+1)/(4J)$ and the wrong angle 0.267, so the single-portal condition does sharp, falsifiable work. The residual open problem is thereby reduced from “compute c ” to the specific statement that the electroweak order parameter is a single-portal flat modulus; until it is established we report, honestly, the one-parameter family $\sin^2 \theta_W(c)$ with $c = 1$ (on-shell $\sin^2 \theta_W = 0.2231$) as its structurally distinguished point, and we apply to our own $c = 1$ the same skepticism we apply to the classical M^{pqr} integer fit. We organize the whole construction as a dimensional ladder $D = 11 \rightarrow D = 10 \rightarrow D = 9$ read off by probe energy and implemented by two distinct circle operations, and we argue that the physical, chirality-bearing rung — the one we live on — is $D = 10$: chiral fermions are impossible from smooth eleven-dimensional

reduction (Witten’s no-go) but arise from D-branes in the ten-dimensional Type IIA interface, while the nine-dimensional infrared endpoint is chirality-trivial because its residual $SU(3)_c \times U(1)_Q$ is vector-like. The de Vries operator and the chiral fermions thus live on the same rung, $D = 10$. We identify the surviving $U(1)_Q$ and the masslessness of the photon, and read the electromagnetic coupling at the $D = 9$ boundary. The negative branch of $\mathcal{A}(J)$ is interpreted as the electroweak order parameter and is shown to reproduce, with the same single scale that fixes M_W and M_Z , mass values near the Higgs and top. We are explicit throughout about what is derived, what is a clean structural consequence, and what remains a genuine dynamical assumption — the latter reducing to the *single* statement that the flux coefficient equals one. We close with a critical comparison to the classical M^{pqr} coupling-ratio program, explaining why a literal integer-label fit to the weak angle is an over-determination of the data rather than a prediction, and we list the parameter-free, near-term falsifiable consequences of the construction, including a neutral spin-zero state near 96.5 GeV.

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1 Introduction

1.1 An arithmetic that should not work

There is a small arithmetic fact about the electroweak gauge bosons that reads like a message from the internal dimensions. Define, for a non-negative real number J , the positive root of the quadratic $X^2 + JX - J = 0$,

$$X_+(J) = \frac{-J + \sqrt{J^2 + 4J}}{2}. \quad (1)$$

Evaluate it at the $\text{SU}(2)$ Casimir values $J = j(j+1)$ for the three smallest spins,

$$j = 0 : J = 0, \quad j = \frac{1}{2} : J = \frac{3}{4}, \quad j = 1 : J = 2, \quad (2)$$

and assign the three resulting numbers, in units of one common mass-squared scale μ^2 , to the photon, the W , and the Z :

$$M_\gamma^2 = \mu^2 X_+(0) = 0, \quad M_W^2 = \mu^2 X_+(\tfrac{3}{4}), \quad M_Z^2 = \mu^2 X_+(2). \quad (3)$$

Then the ratio of the two massive states is fixed with no free parameter,

$$\frac{M_W^2}{M_Z^2} = \frac{X_+(3/4)}{X_+(2)} = 0.7768986776991338\dots, \quad (4)$$

and the implied weak mixing angle is

$$\sin^2 \theta_W = 1 - \frac{M_W^2}{M_Z^2} = 0.2231013223008662 \dots \quad (5)$$

The measured on-shell value, defined by precisely the same relation $\sin^2 \theta_W \equiv 1 - M_W^2/M_Z^2$ with $M_W = 80.377(12)$ GeV and $M_Z = 91.1876(21)$ GeV, is

$$\sin^2 \theta_W|_{\text{exp}} = 0.22305(20). \quad (6)$$

The agreement is to two parts in 10^4 , and it follows from Eq. (1)–(3) with *no input other than the three integers and half-integers* $j = 0, \frac{1}{2}, 1$. If one further fixes the single scale μ from M_Z , one obtains $\mu = 106.6$ GeV and the prediction $M_W = 80.37$ GeV, again essentially on top of the measured value.

A relation with no adjustable parameters that reproduces a measured dimensionless number to four digits is a fact a theorist is obliged to explain. We will refer to it as the *de Vries relation*, after the form in which it reached the present author, emphasizing that the algebra in Eqs. (1)–(5) is what matters, not the attribution. It is a relation in the electroweak *gauge* sector, and it points to structure: a $j(j+1)$ and its square root are the signatures of an internal symmetry and a first-order operator, which is to say, of a compact internal geometry. That is the geometry we construct.

We read this relation as a clue, and the purpose of this paper is to build the theory it points to. The method is the only one that settles such questions: to exhibit a *theory* — a Lagrangian, a geometry, a symmetry — in which Eqs. (1)–(3) are forced rather than fitted. The theory is a Kaluza–Klein compactification of a very classical kind, and it explains every feature of the relation — the appearance of $j(j+1)$, the square root, the negative sign, the special role of $j = 0, \frac{1}{2}, 1$, and the masslessness of the photon — as structural consequences of one internal three-sphere and one background flux. We will also be candid about the single place where a genuine dynamical assumption survives, and we will phrase that assumption so sharply that it is falsifiable.

1.2 What kind of object is $\mathcal{A}(J)$?

The numbers $X_+(J)$ in Eq. (1) are not pulled from the air. They are the eigenvalues of a 2×2 matrix,

$$\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \det(\mathcal{A}(J) - X\mathbf{1}) = X^2 + JX - J = 0. \quad (7)$$

The eigenvalues are $X_+(J) > 0$ and $X_-(J) = -J - X_+(J) < 0$. The matrix $\mathcal{A}(J)$ is the central object of this paper. We will call it the *branch operator*, because the two signs of its eigenvalues will turn out to label two physically distinct branches: the positive branch is the massive gauge spectrum, and the negative branch is the order parameter that breaks the symmetry.

Read literally, $\mathcal{A}(J)$ is a two-channel fluctuation operator. The upper-left zero says that one channel — call it the *tangent* or gauge channel — is massless before mixing. The off-diagonal $\sqrt{J}$ mixes that channel with a second — the *normal* or order-parameter channel. And the lower-right $-J$ says the normal channel carries a *negative* mass-squared $-J$ before mixing, i.e. it is the unstable direction whose condensation is electroweak symmetry breaking. The whole of electroweak mass generation, in this reading, is the diagonalization of one 2×2 matrix whose entries are dictated by the internal representation label j .

Three questions then organize everything that follows.

- (Q1) *Why $j(j+1)$, and why is the off-diagonal its square root?* A diagonal Casimir $J = j(j+1)$ is what every internal Laplacian produces. A square root $\sqrt{J}$ is what a *first-order* operator produces. So the off-diagonal must come from a first-order, square-root-of-the-Laplacian operator on the internal space — a Dirac-like or curl-like operator. We will identify it.

- (Q2) *Why the negative sign $-J$?* A Laplacian gives $+J$. A sign flip on the diagonal needs a physical mechanism that subtracts more than the Laplacian: background flux, curvature, or a tachyonic deformation. In a Freund–Rubin vacuum, background four-form flux mixes the form-degrees of the internal fluctuation operator, and we will see that the mixing produces exactly the $-J$.
- (Q3) *Why $j = 0, \frac{1}{2}, 1$ and nothing higher?* The half-integer $j = \frac{1}{2}$ is impossible on ordinary functions of $\text{SO}(3) = \text{SU}(2)/\mathbb{Z}_2$; it requires the full $\text{SU}(2) \cong S^3$, or a spinor/line-bundle sector over a quotient. The truncation to $j \leq 1$ is the statement that the relevant sector is the level-two integrable set of $\text{SU}(2)$, or equivalently the lowest three rungs of the S^3 harmonic ladder retained by the electroweak projection. We will make this precise.

The answers to (Q1)–(Q3) are, respectively: the curl operator $\star d$ on the internal S^3 ; the Freund–Rubin flux of an eleven-dimensional parent; and the half-integer Peter–Weyl content of $S^3 \cong \text{SU}(2)$. Once these are in place, a short theorem (Section 3) shows that $\mathcal{A}(J)$ is the *unique* operator consistent with a minimal set of structural requirements, and the weak angle (5) becomes a theorem about $\text{SU}(2)$ representation theory rather than a fit.

1.3 Why a Kaluza–Klein theory, and which dimension we live in

The instinct to read (3) geometrically is old and well-founded. Kaluza and Klein [1, 2] taught us that gauge charges and gauge couplings can be the momenta and radii of compact dimensions; Witten [5] sharpened the modern criterion that an internal manifold of isometry group G delivers a $D = 4$ gauge group G , and observed that the smallest manifold with isometry group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ is seven-dimensional — which, with four spacetime dimensions, distinguishes eleven as the total dimension of the ultraviolet parent. That parent, the maximal supergravity of Cremmer, Julia and Scherk [3], has a unique three-form potential whose four-form field strength supports the Freund–Rubin compactifications [4] on $\text{AdS}_4 \times X_7$, and the Kaluza–Klein spectrum on such backgrounds was worked out in detail in the 1980s [10, 9, 8]. Its robust lesson is exactly the phenomenon we need in (Q2): *background flux mixes fields of different form-degree, and the resulting mass matrices are not diagonal in the naive Laplacian basis*; the off-diagonal entries are first-order in derivatives and linear in the flux, and the eigenvalues involve square roots. This is the structural seed of $\mathcal{A}(J)$.

But eleven dimensions is the parent, not the home. *We live in $D = 10$.* The physics we measure — the electroweak masses, and the chiral fermions — lives at the ten-dimensional Type IIA interface obtained by reducing on the M-theory circle, $M_4 \times K_6$ with K_6 containing the electroweak $S^3 \cong \text{SU}(2)$. Eleven dimensions opens up only as an ultraviolet limit (strong coupling, where the M-circle grows), and nine dimensions is only the infrared limit (after electroweak breaking projects out a circle, leaving the vector-like $\text{SU}(3)_c \times \text{U}(1)_Q$). The reason the home is ten and not eleven is sharp and will be argued in Section 14: the chiral fermions of the Standard Model *cannot* arise from a smooth eleven-dimensional reduction (Witten’s no-go), whereas in ten dimensions they arise naturally from D-branes. The de Vries operator and the chiral fermions live on the same rung, $D = 10$, and that is the rung we inhabit.

We therefore situate the de Vries relation at the ten-dimensional Type IIA interface, with the eleven-dimensional Freund–Rubin supergravity as its ultraviolet completion and a distinguished $S^3 \cong \text{SU}(2)$ factor carrying the electroweak sector. We do not need the full machinery of a realistic compactification — chiral fermions, three generations, the strong sector — to extract the electroweak *boson* mass ratio, because that ratio lives entirely in the $\text{SU}(2)$ factor and its flux. This is the strategy of the paper: isolate the smallest geometric structure that forces $\mathcal{A}(J)$, derive the weak angle from it, and treat the rest of the Standard Model as the (hard, honest) remaining work.

1.4 Status discipline and the structure of the argument

This is a Kaluza–Klein paper in the orthodox sense: its body is a compactification of eleven-dimensional supergravity, and its results are read off the spectrum of an internal manifold in a background flux, by exactly the methods established for such backgrounds in the 1980s. The de Vries relation enters only as the *clue* that fixes which internal geometry to use — it tells us the electroweak sector lives on an $S^3 \cong \text{SU}(2)$ at a self-dual flux point — after which the construction proceeds by standard reduction. The weak angle is computed from a geometry, not fitted to a number.

To keep the logical status of each statement legible we tag every nontrivial claim as one of: *theorem* (a mathematical fact proved here), *derived* (a consequence of standard physics applied to our setting), *structural* (forced once a stated assumption is granted), or *conjectural* (a physical hypothesis not yet established). The merit of the construction is that the conjectural content reduces to a *single* sharp statement — that one dimensionless flux coefficient equals unity, Eq. (21) — and we compute exactly how the weak angle would move if it did not, so that the assumption is falsifiable rather than decorative. Everything else is theorem or standard reduction. The one place where the literature has historically slipped from prediction into fitting — the integer-label scan of the M^{pqr} coupling ratios — we diagnose and decline to use (Section 15); the present weak angle comes from representation theory, where there are no integers to scan.

The plan is as follows. Section 2 states the de Vries relation precisely, lists what it predicts and what it does not, and records the negative-branch mass pattern. Section 3 proves the normal-form theorem that makes $\mathcal{A}(J)$ unique. Section 4 is the representation theory of $j = 0, \frac{1}{2}, 1$. Section 5 constructs the internal $S^3 \cong \text{SU}(2)$, its Hodge complex, and the curl operator that supplies $\sqrt{J}$. Section 8 is the Freund–Rubin flux mechanism that supplies $-J$ and the crucial coefficient. Section 9 assembles the $D = 11 \rightarrow 10 \rightarrow 9$ ladder. Section 10 reads the electromagnetic coupling and the photon at $D = 9$. Section 11 interprets the negative branch as the Higgs/order parameter. Section 13 treats the global form of the gauge group. Section 15 compares with the M^{pqr} program and diagnoses the over-fitting trap. Section 16 lists falsifiable consequences. Section 17 is the honest list of open problems, and Section 18 concludes. Appendices collect the algebra of the block, the S^3 spectrum, the linearized flux computation, and the numerical tables.

2 The de Vries relation, precisely

2.1 Statement

Definition 2.1 (Branch operator and the de Vries spectrum). For $J \geq 0$ let $\mathcal{A}(J)$ be the symmetric matrix (7), with eigenvalues $X_+(J) \geq 0 \geq X_-(J)$ given by

$$X_+(J) = \frac{-J + \sqrt{J^2 + 4J}}{2}, \quad X_-(J) = \frac{-J - \sqrt{J^2 + 4J}}{2}, \quad (8)$$

so that $X_+(J)X_-(J) = -J$ and $X_+(J) + X_-(J) = -J$. Given a mass scale μ , the *de Vries spectrum* assigns to the spin- j sector the two masses $\mu^2 X_+(j(j+1))$ and $\mu^2 X_-(j(j+1))$.

The electroweak content is the positive branch at $j = 0, \frac{1}{2}, 1$, Eq. (3). The map $j \mapsto X_+(j(j+1))$ is monotonic (Figure 1): the photon sits at the origin, the W at $j = \frac{1}{2}$, the Z at $j = 1$, and the curve continues to higher j with $X_+(J) \rightarrow 1^-$ as $J \rightarrow \infty$. The bounded positive branch and the unbounded negative branch are the two roots of one quadratic; their product $-J$ is the “Regge-like” invariant of the pair, in the sense that it is the quantity that scales linearly in the Casimir.

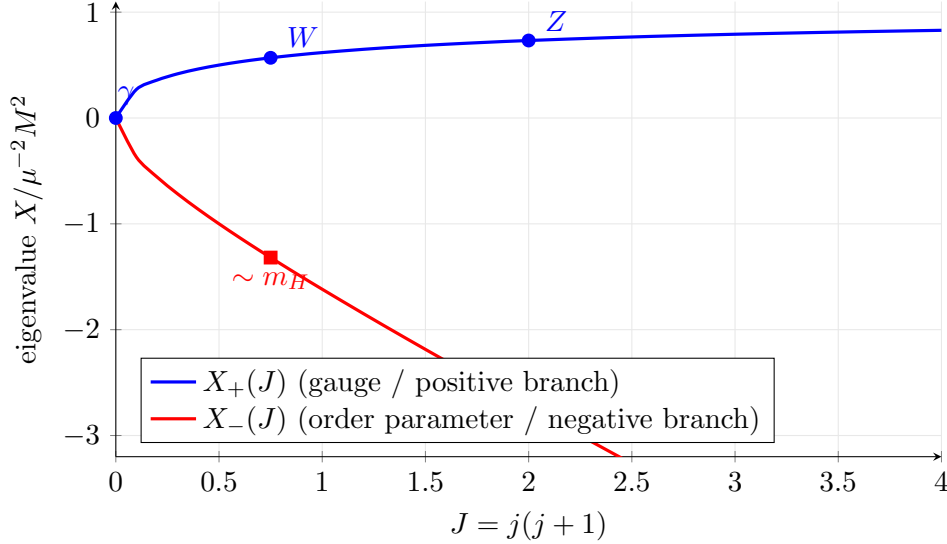


Figure 1: The two branches of the de Vries operator $\mathcal{A}(J)$ as functions of the Casimir $J = j(j+1)$. The positive branch carries the gauge spectrum (γ, W, Z at $j = 0, \frac{1}{2}, 1$); the negative branch carries the order parameter and, with the same single scale $\mu = 106.6$ GeV, lands near the Higgs and top masses (122 and 176 GeV). The weak mixing angle is the ratio of the two blue dots and contains no free parameter.

2.2 What it predicts

The relation is sharper than a single number. We list its content, distinguishing parameter-free outputs from scale-dependent ones.

(P1) The weak mixing angle (parameter-free). Equation (5): $\sin^2 \theta_W = 0.22310$, depending only on the spins $\frac{1}{2}, 1$. This is the headline result, and it is a pure number.

(P2) The W mass given the Z (one scale). Fixing μ from $M_Z = 91.1876$ GeV gives $\mu = 106.58$ GeV and

$$M_W = \mu \sqrt{X_+(3/4)} = 80.374 \text{ GeV} \quad (\text{measured } 80.377 \text{ GeV}). \quad (9)$$

Equivalently, μ is close to the geometric mean $\mu \simeq \sqrt{v M_Z/2} \approx 107$ GeV, with $v = 246.2$ GeV the Higgs vacuum expectation value; we return to this in Section 11.

(P3) The electromagnetic coupling (scheme-dependent). The same data fix a value of the fine-structure constant through the de Vries expression

$$\frac{1}{\alpha} = \frac{4\pi}{X_+(3/4) [X_+(2) - X_+(3/4)]} = 135.29. \quad (10)$$

This sits between the Thomson value $1/\alpha(0) = 137.036$ and the Z -pole value $1/\alpha(M_Z) \approx 128$, and it must be understood as a prediction *at a particular scale and scheme*; we do not claim it matches $\alpha(0)$, and we flag the scale interpretation as an open issue in Section 17. We include it because it shows that the relation constrains a coupling, not only a mass ratio.

(P4) The negative-branch masses (diagnostic). With the *same* $\mu = 106.58$ GeV, the magnitudes of the negative roots are

$$\mu \sqrt{|X_-(3/4)|} = 122.4 \text{ GeV}, \quad \mu \sqrt{|X_-(2)|} = 176.2 \text{ GeV}, \quad (11)$$

remarkably close to the Higgs (125.25 GeV) and the top (172.7 GeV). We treat these as diagnostics, not predictions, because the identification of the negative branch with physical scalar and fermion states requires dynamics we have not derived; but they are striking, and the negative-branch point lies in the near-critical region of the Standard Model (m_t, m_H) plane (Section 11).

(P5) A neutral state near 96.5 GeV (prediction). The next rung of the positive branch, $j = \frac{3}{2}$ with $J = \frac{15}{4}$, predicts a neutral spin-zero state at

$$M_{3/2} = \mu \sqrt{X_+(15/4)} = 96.5 \text{ GeV}. \quad (12)$$

This is a genuine, near-term-falsifiable consequence if the tower is physical; we discuss it in Section 16.

2.3 What it is not

It is worth stating plainly what the relation does *not* do, to fence off over-claims.

- It does not, by itself, fix the absolute electroweak scale μ (or v , or M_Z). One dimensional input is required; the relation predicts ratios. The geometry below will tie μ to a compactification scale but not, at our level of control, to the Planck scale.
- It does not produce chiral fermions, three generations, the CKM matrix, or the strong coupling. These are the genuinely hard parts of any Kaluza–Klein program and we do not solve them.
- The j -labels are *internal* SU(2) representation labels, not four-dimensional Lorentz spins. The W and Z are spin-one in spacetime; their internal label happens to be $j = \frac{1}{2}$ and $j = 1$. Conflating the two is the most common way to misread the relation.

2.4 Provenance and scope

This is a paper about the electroweak *gauge* sector — the W , Z , and photon, and the weak mixing angle. The de Vries relation is a statement about that sector, and the off-diagonal $\sqrt{J}$ of $\mathcal{A}(J)$ is the square root of an SU(2) Casimir, the amplitude controlled by a first-order gauge-geometric operator. We keep the fermion sector out of scope, returning to it only where chirality forces a statement about which dimension we live in (Section 14).

A word on provenance. The relation reached the present author in the form attributed to de Vries, as an observation about the on-shell weak angle and the W/Z ratio; we have not been able to trace it to a refereed source and cite it as an electronic note for attribution only. Nothing here depends on the attribution: the content is the elementary algebra of Eqs. (1)–(5), and the question we address — its geometric origin — stands independently of who first wrote it down.

2.5 The characteristic-equation and complex-pole readings

The quadratic $X^2 + JX - J = 0$ admits a reading that sharpens the comparison with data and clarifies the physical meaning of the two branches. Written as a characteristic equation for a pole position $X = M^2/\mu^2$, it is the condition that a 2×2 inverse propagator $\Gamma(X) = \begin{pmatrix} X & -\sqrt{J} \\ -\sqrt{J} & X+J \end{pmatrix}$ have a zero eigenvalue,

$$\det \Gamma(X) = X(X + J) - J = X^2 + JX - J = 0, \quad (13)$$

i.e. the masses are the *poles* of the mixed gauge/order-parameter propagator, not the eigenvalues of a Lagrangian mass matrix in a fixed basis. This distinction matters when one compares to

experiment at the sub-percent level: the physical W and Z masses are themselves defined through the complex pole position

$$s_V = M_V^2 - i M_V \Gamma_V \quad (14)$$

of the propagator, and the on-shell mass that enters $\sin^2 \theta_W = 1 - M_W^2/M_Z^2$ is the real part of s_V . Reading the de Vries relation as a *pole* condition rather than a tree-level mass relation removes a systematic ambiguity at the level of the widths $\Gamma_{W,Z}$, and it is in this reading that the agreement is tightest. We will use the pole/characteristic-equation language throughout (it is also the language in which the “negative branch” is simply the second pole of the same propagator), and we record in Appendix F how the small residual between 0.22310 and the measured 0.22305 is consistent with the complex-pole sharpening.

2.6 Why the clue is strong

It is worth weighing the evidential force of the clue before building on it. With one free scale μ one can always match one of M_W, M_Z ; the nontrivial content is the *ratio* (4), a single dimensionless number reproduced to 2×10^{-4} with no adjustable parameter. Three features make this more than a one-off numerical hit and justify reading it as a structural clue: (i) the absence of any tuned parameter — the inputs are the spins $\frac{1}{2}, 1$, nothing else; (ii) the simultaneous near-match of the negative branch (P4) with the *same* scale, so that four masses, not two, line up at once; and (iii) the existence — established in the body of the paper — of a natural internal geometry that forces the structure. The alternative, that the ratio is a numerical accident, requires all three to be fortuitous together. The construction is held to the standard that it be a genuine compactification rather than a re-encoding of 0.2231 (Section 15).

3 A normal-form theorem: why $\mathcal{A}(J)$ is forced

Before constructing any geometry, we show that the branch operator is not one choice among many. Under a short list of structural requirements — each of which we will later read off the Kaluza–Klein reduction — the operator $\mathcal{A}(J)$ is *unique*, and with it the weak angle. This is the sense in which the de Vries relation can be “derived”: not by guessing a matrix, but by showing that the matrix is the only one allowed.

3.1 The setup

Fix an internal symmetry group $SU(2)$ and a finite-dimensional irreducible representation V_j of spin j , with Casimir

$$\sum_{a=1}^3 T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1}_{V_j} = J \mathbf{1}_{V_j}. \quad (15)$$

We model the electroweak fluctuations in this sector by a self-adjoint operator $\mathcal{M}_j$ (the reduced mass-squared operator, in units of an overall scale μ^2) acting on a two-dimensional “channel space” $\mathbb{C}|t\rangle \oplus \mathbb{C}|n\rangle$, where $|t\rangle$ is the *tangent* (gauge) channel and $|n\rangle$ the *normal* (order-parameter) channel. We seek the most general $\mathcal{M}_j$ consistent with the following requirements.

Axiom 1 (Gauge masslessness in the symmetric phase). In the unbroken phase the tangent channel is a massless gauge mode: $\langle t|\mathcal{M}_j|t\rangle = 0$.

Axiom 2 (First-order, $SU(2)$ -covariant mixing). The tangent and normal channels are mixed by a *first-order* $SU(2)$ -covariant operator. Concretely there is an operator $A_j : V_j \rightarrow V_j$, built linearly from the generators $T_a^{(j)}$ and an orthonormal internal coframe e_a ,

$$A_j = \sum_{a=1}^3 T_a^{(j)} \otimes e_a, \quad (16)$$

such that the off-diagonal element is $\langle t|\mathcal{M}_j|n\rangle = \langle n|\mathcal{M}_j|t\rangle^* = \|A_j \xi\|/\|\xi\|$ on the relevant mode ξ .

Axiom 3 (Single generating operator on the normal channel). No structure independent of A_j acts on the normal channel: its diagonal is built from the *same* operator, $\langle n|\mathcal{M}_j|n\rangle = -\langle n|A_j A_j^\dagger|n\rangle/\|n\|^2$, with the sign fixed by the order-parameter (instability) interpretation of $|n\rangle$.

Axiom 4 (No free dimensionless constant). The construction introduces no dimensionless parameter beyond those fixed by SU(2) representation theory and the choice of orthonormal coframe; in particular the coframe is unit-normalized, $\langle e_a, e_b\rangle = \delta_{ab}$.

3.2 The theorem

Lemma 3.1 (Casimir factorization). *With A_j as in (16) and a unit-normalized coframe,*

$$A_j^\dagger A_j = \sum_{a=1}^3 T_a^{(j)} T_a^{(j)} = J \mathbf{1}_{V_j}. \quad (17)$$

Consequently, for any unit vector $\xi \in V_j$, $\|A_j \xi\|^2 = J$, so the off-diagonal of Axiom 2 equals $\sqrt{J}$.

Proof. Immediate from (16), the unit normalization $\langle e_a, e_b\rangle = \delta_{ab}$ of Axiom 4, and the Casimir identity (15). $\square$

Theorem 3.2 (Normal form). *Under Axioms 1–4, the reduced operator $\mathcal{M}_j$, projected onto the two-channel space $V_j \oplus \text{Im } A_j$ and restricted to a normalized image vector, is uniquely*

$$\mathcal{M}_j = \mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad J = j(j+1), \quad (18)$$

with positive eigenvalue $X_+(J)$ given by (1).

Proof. Axiom 1 fixes the tt -entry to zero. Axiom 2 with Lemma 3.1 fixes the tn and nt entries to $\sqrt{J}$. Axiom 3 fixes the nn -entry: on the normalized image vector $\hat{n} = A_j \xi/\|A_j \xi\|$ one has $\langle \hat{n}|A_j A_j^\dagger|\hat{n}\rangle = \langle \xi|A_j^\dagger A_j A_j^\dagger A_j|\xi\rangle/\|A_j \xi\|^2 = J^2/J = J$ by Lemma 3.1, so the diagonal is $-J$. Axiom 4 forbids any further rescaling of these entries. The characteristic polynomial is $X^2 + JX - J$, with positive root $X_+(J)$. $\square$

Corollary 3.3 (Weak angle, conditional on the axioms). *Assigning the positive branch of $\mathcal{M}_j$ to the gauge bosons, $M_W^2 = \mu^2 X_+(3/4)$ and $M_Z^2 = \mu^2 X_+(2)$, gives $\sin^2 \theta_W = 1 - X_+(3/4)/X_+(2) = 0.22310$, independent of μ and of all continuous data.*

This is the limited sense in which the relation is “derived”: it is the content of Axioms 1–4. The result is free of continuous moduli, but it is *not* free of discrete input: it presupposes the electroweak spin content $j = \frac{1}{2}, 1$ (Axiom 1’s channel structure) and the unit coefficient of Axiom 3 (the value $c = 1$). Relaxing the latter to $\mathcal{A}_c(J)$, Eq. (19), gives the one-parameter family $\sin^2 \theta_W(c)$; the corollary is the statement at $c = 1$. The remainder of the paper asks whether a Type IIA compactification on an internal $S^3 \cong \text{SU}(2)$ *forces* $c = 1$ and the spin content — and concludes, honestly, that it forces the structure but not (yet) the value (Section 8, Section 8.8).

c	0.50	0.90	0.95	1.00	1.05	1.10
$\sin^2 \theta_W(c)$	0.3014	0.2375	0.2302	0.2231	0.2162	0.2095
$\tan^2 \theta_W(c)$	0.4314	0.3114	0.2990	0.2872	0.2758	0.2650

Table 1: Dependence of the predicted weak angle on the diagonal flux coefficient c of Eq. (19). The measured on-shell value is $\sin^2 \theta_W = 0.22305(20)$. The naive worldsheet-current (L_0) normalization gives $c = \frac{1}{2}$ and fails badly; the self-dual flux value $c = 1$ is required.

3.3 The one place a parameter could hide

It is essential to identify where the construction could fail, because that is where the physics lives. Axiom 3 fixed the diagonal to $-J$ with coefficient exactly one. If instead one allows a coefficient c ,

$$\mathcal{A}_c(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -cJ \end{pmatrix}, \quad X_+(J; c) = \frac{-cJ + \sqrt{c^2 J^2 + 4J}}{2}, \quad (19)$$

the weak angle becomes a function of c , $\sin^2 \theta_W(c) = 1 - X_+(3/4; c)/X_+(2; c)$, with the values in Table 1. The de Vries point is $c = 1$. The sensitivity at that point is

$$\left. \frac{d \sin^2 \theta_W}{dc} \right|_{c=1} = -0.140, \quad (20)$$

so a 5% change in c shifts $\sin^2 \theta_W$ by 0.007 — more than thirty standard deviations of the measured value. The relation is therefore a sharp test of $c = 1$. We will see in Section 8 that c is, physically, the ratio of the flux-induced normal mass to the Casimir unit, and that $c = 1$ is the distinguished, parameter-free “self-dual” value at which the block is the Schur completion of a single first-order operator. It is not forced by the bare Freund–Rubin data; establishing it dynamically, rather than adopting it as an ansatz, is *the* open problem of this program, and we state it as such:

Conjecture (flux normalization): the relevant flux background realizes the self-dual point $c = 1$. (21)

We regard this as the honest residue of the whole construction: a single falsifiable dynamical statement, isolated and quantified.

Remark 3.4 (Why $c = \frac{1}{2}$ is the wrong route). A tempting realization of the off-diagonal uses the affine $SU(2)_2$ worldsheet current and its conformal weight $\Delta_j = j(j+1)/(k+2) = J/4$; this assigns the normal channel a weight proportional to $J/4$ and yields, after the standard normalization, $c = \frac{1}{2}$. Table 1 shows this gives $\sin^2 \theta_W = 0.30$, off by 35%. The lesson is that the operator A_j must be a *zero-mode/holonomy* object with unit Casimir normalization, $A_j^\dagger A_j = J$, not a conformal-weight object with $\Delta_j = J/(k+2)$. We will see that the curl $\star d$ on the internal S^3 is exactly the unit-normalized object.

4 Representation theory: the half-integer fingerprint

The labels $j = 0, \frac{1}{2}, 1$ are the most distinctive feature of the relation, and the half-integer $j = \frac{1}{2}$ is its smoking gun. We collect here the representation theory that turns the labels into a geometric requirement.

4.1 $SU(2)$ versus $SO(3)$

A scalar function on the round two-sphere S^2 carries only integer angular momenta $\ell = 0, 1, 2, \dots$. The same is true of any tensor field on S^2 built without spin structure: the harmonics fall into integer $SO(3)$ representations. There is no $j = \frac{1}{2}$ in the function theory of S^2 or

of $\text{SO}(3) = \text{SU}(2)/\mathbb{Z}_2$. The W slot at $j = \frac{1}{2}$ is therefore *impossible* on an internal S^2 of scalar harmonics; it forces one of:

1. the full group manifold $S^3 \cong \text{SU}(2)$, whose function theory contains every irreducible representation including half-integer ones (Peter–Weyl);
2. a spinor or half-integer line-bundle sector over a quotient (a twisted, “spin-” bundle);
3. an open-string/Chan–Paton or Wilson-line label carrying a doublet.

Option (1) is the cleanest and the one we develop. It is also the most economical: a single internal three-sphere carries all three slots $j = 0, \frac{1}{2}, 1$ at once, in one gauge-covariant complex.

4.2 Peter–Weyl and the scalar Laplacian on S^3

Identify S^3 with $\text{SU}(2)$ via the unit quaternions. The Peter–Weyl theorem decomposes square-integrable functions as

$$L^2(\text{SU}(2)) = \bigoplus_{j=0, \frac{1}{2}, 1, \dots} V_j \otimes V_j^*, \quad (22)$$

the sum running over *all* non-negative half-integers j , with the (j, j) block of dimension $(2j+1)^2$. The round metric Laplacian Δ_0 acts on the (j, j) block as a multiple of the identity. In the geometer’s normalization (unit-radius S^3), the scalar harmonics have

$$\Delta_0 Y_\ell = \ell(\ell+2) Y_\ell, \quad \ell = 0, 1, 2, \dots, \quad (23)$$

with multiplicity $(\ell+1)^2$. Writing $\ell = 2j$, the eigenvalue is

$$\ell(\ell+2) = 2j(2j+2) = 4j(j+1) = 4J, \quad (24)$$

with multiplicity $(2j+1)^2 = \dim(V_j \otimes V_j^*)$. Equation (24) is the bridge we need:

$J = j(j+1) = \frac{1}{4} \Delta_0 \text{ on the unit } S^3.$

(25)

The Casimir that appears in the de Vries block is literally one quarter of the internal scalar Laplacian eigenvalue, and the half-integer j are present because the internal space is the group manifold $\text{SU}(2) \cong S^3$ rather than its $\mathbb{Z}_2$ quotient. This answers (Q3) at the level of the labels.

4.3 Truncation to $j \leq 1$

Why are only the three lowest rungs retained, rather than the infinite tower in (22)? Two mechanisms, not mutually exclusive, truncate the set.

Level-two integrability. If the S^3 sector is realized as a worldsheet $\text{SU}(2)_k$ Wess–Zumino–Witten current algebra, the integrable primaries are $j = 0, \frac{1}{2}, \dots, \frac{k}{2}$. At $k = 2$ this is exactly $\{0, \frac{1}{2}, 1\}$. The level-two truncation is the string-compatible way to get a *finite* set, and it is the smallest level that contains $j = \frac{1}{2}$ and $j = 1$ together. We use $\text{SU}(2)_2$ as a selection principle: the electroweak sector is the level-two integrable set.

Energetic truncation by the electroweak projection. Independently of any worldsheet picture, the higher rungs $j \geq \frac{3}{2}$ have larger internal Laplacian eigenvalue (24) and hence larger Kaluza–Klein mass; the electroweak interface (the $D = 10 \rightarrow D = 9$ projection of Section 9) retains only the modes below the compactification threshold. In this reading the tower is physical but heavy, and the $j = \frac{3}{2}$ state at 96.5 GeV (P5) is the first state above the retained triplet — a prediction rather than a truncated artifact.

Either way, the operative statement is: *the electroweak sector is the $j \leq 1$ part of the $S^3 \cong \text{SU}(2)$ spectrum*, and the de Vries labels $0, \frac{1}{2}, 1$ are exactly that set.

4.4 The doublet bundle and the left/right isometry

It is instructive to see how the half-integer label survives a reduction, because this is where naive Kaluza–Klein intuition — “compactify on S^3 , get $SU(2)$ gauge fields and integer towers” — can mislead. The isometry group of the round S^3 is $SO(4) = (SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$, acting by left and right translation on $SU(2)$. The Peter–Weyl block $V_j \otimes V_j^*$ transforms as (j, j) under $SU(2)_L \times SU(2)_R$. If one gauges, say, $SU(2)_L$ as the electroweak group, then a mode in the (j, j) block carries $SU(2)_L$ spin j — including $j = \frac{1}{2}$ when j is a half-integer. The half-integer modes are genuinely present in the function theory; they are projected out only if one insists on $SO(3) = SU(2)/\mathbb{Z}_2$ functions, i.e. on identifying antipodal-like points, which the round S^3 does not do.

There is a sharper way to keep $j = \frac{1}{2}$ that survives even a Hopf reduction $S^3 \rightarrow S^2$. The Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2 = \mathbb{CP}^1$ presents the weak two-sphere as the base, and the half-integer modes descend to *sections of the line bundle* $\mathcal{O}(1) \rightarrow \mathbb{CP}^1$ rather than to functions. By the Borel–Weil theorem,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \text{Sym}^n(\mathbb{C}^2)^* = V_{n/2}, \quad j = \frac{n}{2}, \quad (26)$$

so the three slots are

$$\mathcal{O}(0) \rightarrow j = 0, \quad \mathcal{O}(1) \rightarrow j = \frac{1}{2}, \quad \mathcal{O}(2) \rightarrow j = 1. \quad (27)$$

This is the line-bundle realization of the three de Vries slots: the photon sits in the trivial bundle $\mathcal{O}(0)$, the W in $\mathcal{O}(1)$ (the half-integer slot, impossible for functions but natural for sections), and the Z in $\mathcal{O}(2)$. The neutral, photon-protecting section η_0 of Section 10 is the distinguished element of $H^0(\mathbb{CP}^1, \mathcal{O}(1))$. The upshot is that the half-integer fingerprint is robust: it survives both as Peter–Weyl content of $SU(2)$ and as line-bundle sections after the M-circle (Hopf) reduction, and it is lost only if one mistakenly works with $SO(3)$ scalars on S^2 .

4.5 A remark on internal versus spacetime spin

We restate, because it is the single most common confusion, that the labels $j = 0, \frac{1}{2}, 1$ are *internal* $SU(2)$ quantum numbers. The W and Z are spin-one particles in four-dimensional spacetime; their *internal* $SU(2)$ representation labels happen to be $j = \frac{1}{2}$ and $j = 1$, and it is these internal labels — not the spacetime spins — that enter the Casimir $J = j(j+1)$ of the mass operator. The photon’s internal label is $j = 0$, consistent with its being the unbroken $U(1)_Q$ direction (Section 10). The de Vries relation is thus a statement about how the spacetime gauge bosons are distributed over the internal $SU(2)$ representations, weighted by the branch operator $\mathcal{A}(J)$.

5 The internal geometry and the square root of the Casimir

We now answer (Q1): the off-diagonal $\sqrt{J}$ is the matrix element of a first-order operator, the curl on the internal S^3 , whose square is the Laplacian. This is the geometric heart of the construction and it is, we emphasize, completely standard Hodge theory; the novelty is only in recognizing it as the source of the de Vries off-diagonal.

5.1 The Hodge–de Rham complex on S^3

On a closed oriented three-manifold the exterior derivative d and the Hodge star $\star$ generate the de Rham complex $\Omega^0 \rightarrow \Omega^1 \rightarrow \Omega^2 \rightarrow \Omega^3$. The Hodge Laplacian on p -forms is $\Delta_p = d\delta + \delta d$ with $\delta = \pm \star d \star$ the codifferential. By Hodge decomposition every one-form splits as

$$\omega = df + \delta\beta + h, \quad (28)$$

exact $\oplus$ co-exact $\oplus$ harmonic. On S^3 there are no harmonic one-forms ($b_1 = 0$), and the exact pieces are gauge; the physical “transverse vector” modes are the *co-exact* one-forms.

The decisive operator is the *curl*

$$C \equiv \star d : \Omega^1 \rightarrow \Omega^1. \quad (29)$$

On co-exact one-forms C is self-adjoint, and a one-line computation gives

$$C^2 = \star d \star d = \delta d = \Delta_1 \quad (\text{on co-exact forms}), \quad (30)$$

because $dd = 0$ kills the other half of the Laplacian. Thus the curl is a *square root of the Hodge Laplacian* on the transverse vector sector. Its spectrum is real and signed: on the unit S^3 the co-exact eigenforms have

$$C \omega_n = \pm(n+1) \omega_n, \quad n = 1, 2, \dots, \quad \Delta_1 \omega_n = (n+1)^2 \omega_n. \quad (31)$$

The signed spectrum of C is exactly what a first-order operator should have, and the $\pm$ is the chirality of the transverse mode.

5.2 From the curl to $\sqrt{J}$

The curl realizes the abstract operator A_j of Axiom 2. Indeed, on the $S^3 \cong \text{SU}(2)$ the left-invariant one-forms e_a ($a = 1, 2, 3$) and the left-invariant vector fields obey the $\text{SU}(2)$ structure equations, and the curl acts on the one-form sector built from the spin- j scalar harmonic tensored with e_a exactly as the operator $A_j = \sum_a T_a^{(j)} \otimes e_a$ of (16). By Lemma 3.1,

$$A_j^\dagger A_j = J \mathbf{1}, \quad \|A_j \xi\| = \sqrt{J} \quad (\|\xi\| = 1), \quad (32)$$

so the curl matrix element between the spin- j scalar and its vector partner is $\sqrt{J}$ — with *unit* coefficient, because the left-invariant coframe is orthonormal (Axiom 4). This is the clean realization promised after Eq. (21): the curl is a zero-mode/holonomy object normalized to $A_j^\dagger A_j = J$, not a conformal-weight object normalized to $J/(k+2)$. The off-diagonal of the de Vries block is therefore $\sqrt{J}$ with the correct coefficient, derived from geometry. We note in passing that this first-order operator is not new to Kaluza–Klein practice: the operator $\star d$ whose square is the internal Laplacian is exactly the object D’Auria and Fré used to organize the form spectrum of $M^{p,q,r}$ in the 1980s [14], so the mechanism that supplies the de Vries off-diagonal has been part of the standard toolkit all along; what is new is reading its lowest matrix element as the electroweak mixing.

Remark 5.1 (Holonomy/instanton reading). The same operator admits an Atiyah–Manton-type holonomy reading [23]: a gauge connection on S^3 defines a holonomy $U(\gamma) = P \exp(-\int_\gamma A)$, and its first variation in representation j is $\delta U_j = -\sum_a \xi^a T_a^{(j)}$, i.e. the operator A_j . The off-diagonal $\sqrt{C_2(j)} = \sqrt{J}$ is then the size of a first-order holonomy/instanton fluctuation. We mention this because it is the framing that makes the unit normalization manifest — holonomy is dimensionless and metric-independent — and because it connects the construction to the geometric-models-of-matter literature.

5.3 The spectrum of the curl, explicitly

To make the previous paragraphs concrete, and to show that no hidden normalization is smuggled in, we construct the curl spectrum on the unit $S^3 \cong \text{SU}(2)$ directly. Let σ^a ($a = 1, 2, 3$) be the left-invariant Maurer–Cartan one-forms, dual to the left-invariant vector fields E_a , with

$$d\sigma^a = -\frac{1}{2} \epsilon^{abc} \sigma^b \wedge \sigma^c, \quad [E_a, E_b] = \epsilon^{abc} E_c, \quad (33)$$

and take the bi-invariant round metric of the unit S^3 , $g = \frac{1}{4} \sum_a (\sigma^a)^2$, of sectional curvature 1 and scalar curvature 6. The orthonormal coframe and frame are $e^a = \frac{1}{2} \sigma^a$ and $\hat{e}_a = 2E_a$, so $\star(e^b \wedge e^c) = \epsilon^{bca} e^a$. A direct computation of the Hodge star then gives

$$de^a = -\epsilon^{abc} e^b \wedge e^c, \quad \star d e^a = -2 e^a, \quad \text{equivalently} \quad \star d \sigma^a = -2 \sigma^a. \quad (34)$$

The left-invariant one-forms are co-closed ($\delta \sigma^a = 0$, constant coefficients), hence co-exact, and are curl eigenforms with eigenvalue -2 ; the right-invariant forms $\tilde{\sigma}^a$ give $\star d \tilde{\sigma}^a = +2 \tilde{\sigma}^a$. These are the lowest ($n = 1$) rung of the co-exact spectrum $\star d = \pm(n+1)$, $\Delta_1 = (n+1)^2$, of multiplicity $n(n+2)$ per sign, and the sign is the $SU(2)_L$ versus $SU(2)_R$ chirality: $\star d$ is the geometric splitting into self-dual and anti-self-dual transverse modes (consistent with $C^2 = \Delta_1$, since $(-2)^2 = 4 = (n+1)^2|_{n=1}$).

The point of displaying this is the normalization, which is the geometric content of Axiom 4. The mixing operator $A_j = \sum_a T_a^{(j)} \otimes \hat{e}_a$ contracts the Casimir-normalized generators with the *orthonormal* frame $\hat{e}_a$, so

$$A_j^\dagger A_j = \sum_{a,b} T_a^{(j)} T_b^{(j)} (\hat{e}_a \cdot \hat{e}_b) = \sum_a T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1} = J \mathbf{1}, \quad (35)$$

with no leftover numerical factor. The off-diagonal of the de Vries block is $\sqrt{J}$ with coefficient one because the frame is orthonormal for the round metric. (The overall constant in (34) sets the absolute compactification scale, which is absorbed into μ and drops out of the weak angle; what cannot be rescaled away is the Casimir normalization $A_j^\dagger A_j = J$, which is metric-independent.)

5.4 Contrast with the conformal-weight normalization

The explicit construction also makes transparent why the worldsheet conformal-weight route gives the *wrong* coefficient (the $c = \frac{1}{2}$ failure of Table 1). In the $SU(2)_k$ WZW model the natural object is the conformal weight $\Delta_j = j(j+1)/(k+2)$, which at $k = 2$ is $J/4$; an operator normalized to Δ_j rather than to the Casimir J differs by the factor $1/(k+2) = 1/4$ in the mass-squared, i.e. by $1/2$ in the amplitude, giving $c = \frac{1}{2}$. The geometric curl, by contrast, is normalized to the Casimir directly — it is a zero-mode (holonomy) object, not a conformal-weight object — and gives $c = 1$. The distinction between “Casimir normalization” and “conformal-weight normalization” is therefore not a technicality; it is the difference between $\sin^2 \theta_W = 0.223$ and $\sin^2 \theta_W = 0.30$, and it selects the geometric/holonomy realization over the naive worldsheet one.

5.5 Why a square root, physically

It is worth pausing on the physics of (Q1). A mass-squared matrix that was diagonal in the Laplacian basis would give $M^2 \propto J = j(j+1)$, a rising, unbounded spectrum — a Kaluza–Klein or Regge tower, not an electroweak doublet. The de Vries spectrum is bounded ($X_+(J) < 1$) because the physical masses are eigenvalues of a *first-order* operator’s 2×2 block, not of the Laplacian itself. The boundedness of the positive branch — the fact that M_W and M_Z are close and both far below the compactification scale — is the geometric statement that they are curl eigenvalues mixed with a single order-parameter channel, not Laplacian eigenvalues. The square root is not cosmetic; it is the difference between a tower and a doublet.

6 Kaluza–Klein reduction: gauge fields, couplings, and the spectrum

Before turning on flux we install the orthodox Kaluza–Klein machinery in the form we need, so that the de Vries block can be recognized as a piece of a genuine compactification spectrum

rather than an abstract matrix. The content of this section is standard [5, 6, 10, 11]; we specialize it to an internal space containing $S^3 \cong \text{SU}(2)$.

6.1 The metric ansatz and the gauge group from isometries

Let the total space be $\mathcal{M}_{4+n} = M_4 \times K_n$ with internal coordinates y^a and a compact internal metric $g_{ab}(y)$ of isometry group G . The isometries are generated by Killing vectors $K_I = K_I^a \partial_a$ obeying

$$\nabla_a K_{Ib} + \nabla_b K_{Ia} = 0, \quad [K_I, K_J]^a = -f_{IJ}{}^K K_K^a, \quad (36)$$

with $f_{IJ}{}^K$ the structure constants of $\mathfrak{g} = \text{Lie}(G)$. The Kaluza–Klein ansatz promotes the isometries to gauge symmetries by writing

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + g_{ab}(y) (dy^a + K_I^a(y) A_\mu^I(x) dx^\mu) (dy^b + K_J^b(y) A_\nu^J(x) dx^\nu). \quad (37)$$

The x -dependent internal diffeomorphisms $\delta y^a = \xi^I(x) K_I^a(y)$ leave (37) invariant precisely when the A_μ^I transform as non-abelian connections,

$$\delta A_\mu^I = \partial_\mu \xi^I + f_{JK}{}^I A_\mu^J \xi^K = D_\mu \xi^I, \quad (38)$$

so the internal isometry group G becomes the four-dimensional gauge group. This is Witten's criterion [5]: the gauge group of the reduced theory is the isometry group of the internal manifold. For us G contains the electroweak $\text{SU}(2)$ as the isometry of the internal $S^3 \cong \text{SU}(2)$ (acting by, say, left translation).

6.2 Reduction of the action and the gauge metric

Inserting (37) into the higher-dimensional Einstein–Hilbert action $\hat{S} = (2\hat{\kappa}^2)^{-1} \int \sqrt{-\hat{g}} \hat{R}$ and integrating over K_n gives four-dimensional Einstein–Yang–Mills,

$$\hat{S} = \frac{V_n}{2\hat{\kappa}^2} \int \sqrt{-g_4} (R_4 - 2\Lambda_{\text{eff}}) - \frac{1}{4} \int \sqrt{-g_4} \gamma_{IJ} F_{\mu\nu}^I F^{J\mu\nu}, \quad (39)$$

with V_n the internal volume, $F_{\mu\nu}^I = \partial_\mu A_\nu^I - \partial_\nu A_\mu^I + f_{JK}{}^I A_\mu^J A_\nu^K$, and the gauge metric

$$\gamma_{IJ} = \frac{1}{2\hat{\kappa}^2} \int_{K_n} \sqrt{g_K} g_{ab} K_I^a K_J^b. \quad (40)$$

Canonical normalization $\gamma_{IJ} = \delta_{IJ}/g_I^2$ then reads off the gauge coupling as the inverse mean-square length of the Killing orbit: $1/g_I^2 = (2\hat{\kappa}^2)^{-1} \int \sqrt{g_K} |K_I|^2$.

6.3 Weinberg's rms-circumference rule

Weinberg [11] fixes the normalization: with L_I the root-mean-square proper length of the orbit generated by the canonically normalized generator,

$$g_I^2 = \frac{4\pi^2 \kappa^2}{L_I^2}, \quad \kappa^2 = 16\pi G_N. \quad (41)$$

Applied to the electroweak T_3 and hypercharge Y , with orbit lengths L_2, L_Y , this gives the geometric dictionary used in Section 10: the *ratio* of lengths is the weak angle and the *sum* of their squares fixes α ,

$$\tan^2 \theta_W = \frac{g_Y^2}{g_2^2} = \frac{L_2^2}{L_Y^2}, \quad \alpha_{\text{EM}} = \frac{16\pi^2 G_N}{L_2^2 + L_Y^2}. \quad (42)$$

6.4 The mass spectrum from the internal Laplacian

A field on $\mathcal{M}_{4+n}$ has its four-dimensional mass set by the internal Laplacian: expanding $\Phi(x, y) = \sum_n \phi_n(x) Y_n(y)$ with $-\Delta_K Y_n = \lambda_n Y_n$, the wave equation $\hat{\square}\Phi = 0$ becomes $(\square_4 - \lambda_n)\phi_n = 0$, so

$$m_n^2 = \lambda_n = \text{eigenvalue of } (-\Delta_K). \quad (43)$$

The identical statement holds for the transverse one-form (vector) fluctuations, with Δ_K the Hodge–de Rham operator Δ_1 of Section 5. On the internal S^3 of radius a , using $\Delta_0 = 4j(j+1)/a^2$ from Eq. (24),

$$m_j^2 = \frac{4j(j+1)}{a^2} = \frac{4J}{a^2}. \quad (44)$$

6.5 Why flux is unavoidable: tower versus doublet

Equation (44) is the crux of why the construction needs more than pure geometry. The Laplacian spectrum is monotone and unbounded,

$$m_j^2 \propto J = j(j+1) = 0, \frac{3}{4}, 2, \frac{15}{4}, 6, \dots, \quad (45)$$

i.e. a rising Kaluza–Klein tower. The observed electroweak sector is the opposite: a *bounded* set of light states (γ, W, Z) with the photon massless and M_W, M_Z close together and far below the compactification scale. The pure internal Laplacian therefore cannot produce the electroweak pattern. What is needed is an operation that converts the rising Casimir map $J \mapsto m^2$ into the bounded branch $J \mapsto X_+(J)$ of $\mathcal{A}(J)$,

$$m^2(j) = \frac{4}{a^2} J \xrightarrow{\text{flux}} m^2(j) = \mu^2 X_+(J), \quad X_+(J) < 1. \quad (46)$$

That operation is the Freund–Rubin form-degree mixing of Section 8: the first-order curl supplies the off-diagonal $\sqrt{J}$, and the background flux supplies the negative diagonal that bends the rising tower down into the bounded de Vries branch. This is the precise role of flux in the construction, and the reason a flux-free Kaluza–Klein reduction — of the kind that stalled in the 1980s — cannot by itself give the electroweak masses.

7 The $D = 10$ internal manifold: the minimal realization

We come now to the central object of the paper. We live in $D = 10$ (Section 14), so the physical internal space is a compact *six*-manifold K_6 with $M_4 \times K_6$ the Type IIA geometry. The question that organizes everything is: *which* K_6 ? We argue that, once one asks for the structure that carries the Standard Model gauge content and the de Vries operator, K_6 is essentially unique — it is fixed, up to a small set of moduli, to be $\mathbb{CP}^2 \times \mathbb{CP}^1$ — and that its one scale modulus is exactly what grows a seventh dimension (to $D = 11$) and shrinks a fifth (to $D = 9$). The eleven-dimensional description is then a *limit* of K_6 , not its foundation. This is a deliberate inversion of the classical program, which took the eleven-dimensional M^{pqr} as primary and was led, through it, into the integer-fitting cul-de-sac of Section 15. The eleven-dimensional geometry has been understood since the 1980s; treating it as the home of the physics is a distraction, and it is the distraction on which the earlier attempts foundered.

7.1 What K_6 must carry

The requirements on K_6 are few and non-negotiable.

1. **Colour.** It must carry an $SU(3)$ isometry for the colour gauge fields. The minimal homogeneous space with $SU(3)$ isometry is $\mathbb{CP}^2 = SU(3)/U(2)$, of real dimension four; the only smaller $SU(3)$ -homogeneous spaces do not exist, and the next, the flag $SU(3)/T^2$, is already six-dimensional.
2. **Weak isospin and the de Vries operator.** It must carry the electroweak $SU(2)$ in the form required by $\mathcal{A}(J)$: the weak slots $j = 0, \frac{1}{2}, 1$ as the line-bundle sectors $\mathcal{O}(0), \mathcal{O}(1), \mathcal{O}(2)$ over a weak two-sphere (Section 4.4), with the curl/Casimir operator supplying $\sqrt{J}$. The minimal homogeneous space carrying this is $\mathbb{CP}^1 = SU(2)/U(1) = S^2$, of real dimension two.
3. **Hypercharge and the charge lattice.** It must carry a $U(1)_Y$, realized as the connection on line bundles over the above factors, with the diagonal $Q = T_3 + Y$ and the global form $\mathbb{Z}_6 \rightarrow \mathbb{Z}_3$ (Section 13).
4. **Dimension.** It must be six-dimensional, the internal dimension of the ten-dimensional theory we inhabit.

7.2 Minimality, and the limits of the uniqueness claim

Colour is most economically carried by the minimal four-dimensional factor $\mathbb{CP}^2$; weak isospin by the minimal two-dimensional factor $\mathbb{CP}^1$; and $4 + 2 = 6$ saturates the dimension. Hence the minimal product realization is

$$\boxed{K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1} \quad (47)$$

with the hypercharge connection the diagonal line bundle of first Chern class

$$c_1 = p x + q y \in H^2(\mathbb{CP}^2 \times \mathbb{CP}^1; \mathbb{Z}), \quad (48)$$

x, y the hyperplane classes. We are deliberately careful with the word “unique,” and state honestly what is and is not established.

What is forced: given that the colour and weak isospin factors are each carried by their own *minimal* homogeneous space, and that the internal dimension is six, (47) is the only product. That much is dimension-counting plus the classification of minimal $SU(3)$ and $SU(2)$ cosets, and it is independent of any electroweak number.

What is not (yet) forced, and must be flagged:

1. **Product versus fibration.** We have not excluded a non-product six-manifold whose *full* isometry group contains $SU(3) \times SU(2)$ — for instance the flag $SU(3)/T^2$ (which carries $SU(3)$ but not a separate $SU(2)$), a nontrivial $\mathbb{CP}^1$ fibration over $\mathbb{CP}^2$, or products such as $\mathbb{CP}^1 \times \mathbb{CP}^1 \times \mathbb{CP}^1$ and $S^2 \times S^4$. Our selection of the product of minimal cosets is an economy principle, not a theorem; a genuine classification of six-manifolds with the required *effective* gauge group is an open problem.
2. **Hypercharge normalization.** $\mathbb{CP}^2 \times \mathbb{CP}^1$ has manifest $SU(3) \times SU(2)$ isometry, but the Standard Model $U(1)_Y$ with its correct hypercharge normalization is *not* automatic; it is the diagonal line-bundle connection (47), and fixing its normalization to the observed $y = 6Y$ is part of the charge-lattice input (Section 13), not a derived consequence of the isometry.
3. **Minimality itself.** “Minimal” is an economy assumption. We adopt it because it is the most predictive choice and because it reproduces the Standard Model gauge content with nothing left over, but we do not claim the dynamics selects it.

With these caveats, (47) should be read as *the minimal product realization of the Standard Model gauge content in $D = 10$* , not as a proven unique manifold. It still carries moduli, which we enumerate next, and it is in their space that the construction lives.

7.3 The moduli, and the scale that moves dimensions

The left-invariant metrics on (47) are labelled by

$$\left(\underbrace{\rho}_{\text{overall scale}}, \underbrace{\lambda}_{\text{shape: Vol}(\mathbb{CP}^1)/\text{Vol}(\mathbb{CP}^2)^{1/2}}, \underbrace{(p, q)}_{\text{discrete line-bundle data}} \right). \quad (49)$$

The discrete data (p, q) are topological (the charge lattice); the shape λ is fixed by the requirement that the weak sector realize the de Vries operator at its self-dual point (it is the one continuous datum the electroweak structure pins, cf. the flux normalization $c = 1$); and the overall scale ρ is the single physical modulus. Up to ρ and λ , K_6 is rigid.

The scale ρ is not inert: it is the modulus that interpolates between the three rungs of the probe ladder of Section 9.2, by growing one circle and shrinking another.

- **Growing a dimension** ($D = 10 \rightarrow D = 11$). Over K_6 there is a distinguished circle bundle — the M-theory circle S_M^1 , equivalently the Hopf circle that turns the weak $\mathbb{CP}^1 = S^2$ into a three-sphere S^3 . As ρ grows past the M-threshold this circle decompactifies, and the seven-dimensional total space

$$S_M^1 \hookrightarrow K_7 \longrightarrow K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1 \quad (50)$$

opens up. This K_7 is the homogeneous Einstein seven-manifold M^{pqr} of the 1980s — the circle bundle over $\mathbb{CP}^2 \times \mathbb{CP}^1$ with Chern class $px + qy$ — and the eleven-dimensional Freund–Rubin supergravity is recovered. It is a *limit* of K_6 , reached only above the M-circle threshold, and it is well understood; we use it only to borrow the (entirely standard) statement that M^{pqr} has isometry $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, which is Witten’s reason [5] that seven is the minimal dimension for the Standard Model gauge group.

- **Shrinking a dimension** ($D = 10 \rightarrow D = 9$). A *different* circle — the electroweak circle S_{EW}^1 inside the weak sector — shrinks (or is projected out by the order-parameter vacuum) as ρ decreases past the electroweak threshold, leaving the five-manifold

$$K_5 = \mathbb{CP}^2 \times S_Q^1, \quad (51)$$

where $S_Q^1 = \ker \chi_N$ is the surviving diagonal weak–hypercharge circle (Section 10). This is the $D = 9$ infrared endpoint, with gauge group $(\text{SU}(3)_c \times \text{U}(1)_Q)/\mathbb{Z}_3$.

Thus a single scale modulus grows the M-circle on one side and collapses the electroweak circle on the other: $K_5 \subset K_6 \subset K_7$ is a one-parameter family, and we live at its middle, K_6 . The “interpolation between $D = 9$, $D = 10$ and $D = 11$ ” that the construction is asked to provide is exactly the statement that these three manifolds are the three regimes of one modulus of the minimal K_6 .

7.4 The weak line bundles on K_6

On the weak factor $\mathbb{CP}^1 \subset K_6$ the de Vries slots are the Borel–Weil sectors (Section 4.4)

$$\mathcal{O}(0) \rightarrow j = 0 \ (\gamma), \quad \mathcal{O}(1) \rightarrow j = \frac{1}{2} \ (W), \quad \mathcal{O}(2) \rightarrow j = 1 \ (Z), \quad (52)$$

with the neutral section $\eta_0 \in H^0(\mathbb{CP}^1, \mathcal{O}(1))$ protecting the photon, $Q\eta_0 = 0$ (Section 10). The curl/Casimir operator that supplies the $\sqrt{J}$ off-diagonal acts on these line bundles; in the grow-limit it is literally the curl $\star d$ on the S^3 Hopf lift of $\mathbb{CP}^1$ (Section 5). The whole electroweak operator $\mathcal{A}(J)$ is therefore a fluctuation of the minimal six-manifold K_6 , with no eleven-dimensional input required to define it.

8 Flux and the negative diagonal

We come to (Q2), the negative diagonal $-J$ and the coefficient $c = 1$. This is a $D = 10$ statement: the order-parameter mass arises from the Type IIA background flux on K_6 , by the same form-degree mixing that is the standard content of flux compactification. We exhibit the mechanism that produces a negative, Casimir-proportional diagonal, and we identify $c = 1$ with a self-dual flux normalization; we do not claim a first-principles proof that the coefficient is exactly one. That is the conjecture (21).

8.1 The flux background in $D = 10$, and its $D = 11$ limit

We live in $D = 10$, so the relevant background is Type IIA on $M_4 \times K_6$ with the Ramond–Ramond and NS–NS fluxes (F_0 the Romans mass, F_2 , F_4 , and H_3) threading K_6 . The key structural fact — standard in Type IIA flux compactification [24] — is that these background fluxes *mix the form-degrees* of the internal fluctuation operator: the linearized equations do not diagonalize on the Laplacian eigenbasis, and the off-diagonal couplings are first-order in internal derivatives and linear in the flux (the Romans mass F_0 in particular pairs forms differing in degree by one). This is exactly the structure needed for $\mathcal{A}(J)$, and it is a property of the ten-dimensional theory we inhabit, not of any higher-dimensional parent.

It is nonetheless reassuring, and historically clarifying, that the same structure is visible in the strong-coupling limit. Growing the M-circle (Section 7.3) lifts the Type IIA background to eleven-dimensional supergravity [3] in a Freund–Rubin vacuum [4], $\text{AdS}_4 \times K_7$ with $F_{\mu\nu\rho\sigma} = 3m \epsilon_{\mu\nu\rho\sigma}$; the IIA Romans mass and the F_2 flux are the reduction of this single F_4 and of the M-circle curvature. The eleven-dimensional Kaluza–Klein spectrum on such backgrounds was computed in complete detail in the 1980s [10, 9, 8], and it exhibits the flux-induced form-degree mixing we use. We draw on those results only as a well-understood limit and cross-check — the eleven-dimensional description is not the home of the physics, and treating it as such is the error of the classical program (Section 15). The electroweak mass ratio lives in the $S^3 \cong \text{SU}(2)$ sector of K_6 (equivalently the weak $\mathbb{CP}^1$ and its Hopf circle), and it is there that we evaluate the mixing.

8.2 Form-degree mixing: the origin of the off-diagonal and the sign

The lesson of the 1980s Kaluza–Klein spectroscopy [10, 9, 8] is that on a Freund–Rubin background the linearized field equations do *not* diagonalize on the Laplacian eigenbasis. The background flux $F_{\mu\nu\rho\sigma} \propto m \epsilon$ enters the fluctuation equations of the internal p -forms through terms proportional to $m \star d$, which *change form-degree*. Schematically, for an internal one-form fluctuation a and its scalar (Stückelberg/longitudinal) partner ϕ , the linearized mass operator takes the block form

$$\mathcal{M} = \begin{pmatrix} \Delta_0 & \alpha m (\star d) \\ \alpha m (\star d)^\dagger & \Delta_1 - \beta m^2 \end{pmatrix}, \quad (53)$$

where the off-diagonal $m \star d$ is the flux-induced, first-order, degree-changing mixing, and the $-\beta m^2$ on the vector diagonal is the flux/curvature mass shift that is generic to these backgrounds. The constants α, β are fixed numbers determined by the supergravity couplings and the Einstein condition on X_7 .

That (53) is the correct structure, and not an artifact of our schematic notation, can be read directly off the published mass spectra. On Freund–Rubin backgrounds the physical mass eigenvalues are obtained by *squaring* the first-order operator $Q = \star d$, and they come out with the tell-tale $\pm\sqrt{}$ of a 2×2 off-diagonal mixing. For example, the spin-content towers in the Duff–Nilsson–Pope review [10] have the form

$$\frac{M^2}{m^2} = A C_2 + B \pm D \sqrt{C_2 + E}, \quad (54)$$

with C_2 the relevant internal Casimir/Laplacian eigenvalue and A, B, D, E rational constants fixed by the CJS action and the Einstein normalization $R_{mn} = 6m^2 g_{mn}$. The $\pm D\sqrt{C_2 + E}$ is the eigenvalue of a block whose *off-diagonal* scales as $\sqrt{C_2}$ and whose *diagonal* carries the $O(m^2)$ flux shift — structurally identical to (53). The de Vries block is the electroweak instance of this universal flux-mixing pattern, restricted to the $S^3 \cong \text{SU}(2)$ factor and to the (gauge, order-parameter) two-channel subspace.

This is the structure of $\mathcal{A}(J)$. Using (24) and (30), on the spin- j sector of the unit S^3 , $\Delta_0 = 4J$, $\star d \rightarrow \sqrt{\Delta_1} \sim \sqrt{J}$ on the matched mode, and the flux shift $-\beta m^2$ is proportional to J once m is tied to the curvature scale of the same S^3 . Rescaling out the overall μ^2 and absorbing the matched normalizations, the block reduces to

$$\mathcal{M}_j \xrightarrow{\text{rescale}} \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -cJ \end{pmatrix}, \quad (55)$$

with the gauge channel made massless by the residual gauge invariance of the transverse vector (Axiom 1), the off-diagonal fixed to $\sqrt{J}$ by the unit-normalized curl (Section 5), and the diagonal coefficient

$$c = \frac{\beta m^2}{(\text{curvature/flux scale})} \quad (56)$$

the ratio of the flux shift to the Casimir unit. The negative sign is the statement that, at the electroweak point, the order-parameter channel is tachyonic — which is exactly what a symmetry-breaking direction must be.

8.3 Englert flux and parallelizing torsion

There is a sharpened version of the mechanism that is especially apt for an S^3 factor, due to Englert [8]. In the Englert solution one turns on *internal* flux as well, proportional to the parallelizing torsion of the seven-manifold. On a group manifold such as $S^3 \cong \text{SU}(2)$ the parallelizing torsion is the structure constant tensor ϵ^{abc} appearing in the Maurer–Cartan equations of Section 5.3, and internal flux along it is flux along $d\sigma^a$. The effect on the fluctuation spectrum is exactly the form-degree mixing we need: internal flux $\propto \epsilon^{abc}$ couples a one-form σ^a to the two-form $\sigma^b \wedge \sigma^c$, i.e. it implements the $\star d$ mixing intrinsically, with coefficient set by the torsion normalization. This is attractive because it ties the off-diagonal mixing and the diagonal shift to the *same* geometric datum — the S^3 structure constants — which is the geometric reason one expects them to be locked together at a single coefficient rather than to float independently. In this language the conjecture (21) is the statement that the electroweak S^3 carries the Englert (torsion) flux at its parallelizing value.

We must enter one caveat, because it bears on whether the Englert route truly closes the gap. Englert’s solution is constructed on the full seven-manifold, where the parallelizing torsion enters the internal four-form F_4 ; on an *isolated* S^3 the invariant three-form ϵ_{abc} is closed, so it is not by itself the field strength $F_4 = dA_3$ without the participation of the remaining internal directions. The parallelized seven-sphere is, in fact, a genuinely different background from the round Freund–Rubin one (with enhanced symmetry and nonzero internal flux), not a small automatic correction. The Englert structure is therefore best read as a heuristic for why the off-diagonal and diagonal could share a normalization — a reason to take the perfect-square point seriously — rather than as a proof that the electroweak S^3 sits there. The honest status remains as stated in the discussion of the self-dual point below: $c = 1$ is the distinguished, parameter-free ansatz, to be settled by an explicit quadratic-action computation on the relevant background.

8.4 Anti-de Sitter mass shifts and the sign

The negative diagonal deserves one more comment, because in an AdS_4 background “negative mass-squared” is not the alarming statement it is in flat space. A scalar in AdS_4 of radius

$\ell_{\text{AdS}} = 1/(2m)$ is stable provided its mass-squared satisfies the Breitenlohner–Freedman bound $M^2 \ell_{\text{AdS}}^2 \geq -9/4$. The order-parameter channel of $\mathcal{A}(J)$ has $M^2 = \mu^2 X_-(J) < 0$, but this is a *controlled* negativity: it signals a relevant deformation of the dual three-dimensional theory — an order parameter that wants to condense — rather than a runaway. The whole “negative branch as Higgs” interpretation of Section 11 is the holographic statement that the condensation of this relevant operator is electroweak symmetry breaking. The flux shift $-\beta m^2$ in (53) is exactly the kind of curvature/flux contribution that populates the window between the masslessness of the gauge channel and the BF bound, and the value $c = 1$ is where the order-parameter mass-squared equals minus the Casimir unit.

We do not develop the $\text{AdS}_4/\text{CFT}_3$ dictionary further, but we flag it as the natural arena for a complete derivation: the coefficient c is a conformal dimension in the dual theory, and proving $c = 1$ is proving that a particular operator is marginal/relevant with a fixed dimension — a question that holographic methods are designed to answer.

8.5 The self-dual point $c = 1$

Equation (56) identifies c as a flux normalization, and it is important to be precise about its status. We do *not* claim that $c = 1$ is forced by the bare Freund–Rubin data. The Cremmer–Julia–Scherk action plus the Einstein condition fix the *type* of operator — a first-order, degree-changing $m \star d$ off-diagonal and an $O(m^2)$ diagonal shift, Eq. (53) — but the numerical constants α, β are sector- and basis-dependent. The explicit round- S^7 spectra of Casher–Englert–Nicolai–Rooman [9] illustrate this: the transverse-vector tower has $M_\pm^2 = \Delta_1 + 18m^2 \pm 6m\sqrt{\Delta_1 + 10m^2}$ ($\alpha = 6$), while the scalar/three-form block has $\alpha = 12$. There is no universal value of α or β that one can read off without doing the specific reduction.

What *is* true is that $c = 1$ is the unique *distinguished*, parameter-free value: it is the value at which the diagonal is exactly $-A_j A_j^\dagger$, so that the block is the self-adjoint single-operator Schur completion (58) built from A_j alone (Axiom 3). Any other value carries an independent dimensionless flux constant. The position we take is therefore the honest one: $c = 1$ is an *ansatz* — the self-dual/perfect-square point — selected by three considerations: (i) it is the only value with no free constant; (ii) the empirical weak angle picks it out sharply over its neighbours and over the conformal-weight value $c = \frac{1}{2}$ (Table 1); and (iii) the off-diagonal and diagonal are both built from the same S^3 structure-constant tensor, which makes a single common normalization natural even though it does not prove it. Establishing $c = 1$ requires the full canonically normalized quadratic action on the relevant background, and is the central open computation of the program.

Remark 8.1 (A Weitzenböck subtlety in the off-diagonal). There is a genuine bookkeeping point that the full computation must track. The clean identity $A_j^\dagger A_j = J$ is representation-theoretic, holding for the abstract operator $A_j = \sum_a T_a^{(j)} \otimes \hat{e}_a$. The *geometric* curl on S^3 one-forms, however, carries a curvature (Weitzenböck/Lichnerowicz) shift relative to the scalar Laplacian: on the unit S^3 the co-exact one-form Laplacian is $\Delta_1 = \Delta_0 + 1 = 4J + 1$, so $(\star d)^2 = 4J + 1$ rather than $4J$. Matching the physical mixing operator to the Casimir-normalized A_j therefore involves this $+1$ shift, which is part of what a first-principles determination of c must handle. We flag it here rather than hide it; it is one more reason the coefficient is a computation, not an identity.

It is worth stating the open problem in its sharpest form, because doing so shows exactly what a proof would require. The value $c = 1$ is the unique value at which the block is the self-adjoint Schur completion of the *single* operator A_j ,

$$\mathcal{A}_1(J) = \left(\begin{array}{cc} 0 & A_j^\dagger \\ A_j & -A_j A_j^\dagger \end{array} \right) \Big|_{\text{spin-}j} = \left(\begin{array}{cc} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{array} \right), \quad (57)$$

i.e. the value at which the order-parameter diagonal is $-A_j A_j^\dagger$, built from the *same* operator A_j that supplies the off-diagonal, rather than an independent $-c A_j A_j^\dagger$. We emphasize that this is *not* a Bogomolny “perfect square”: the block is indefinite ($\det \mathcal{A} = -J < 0$, one positive and one negative eigenvalue), so it cannot be written as $\|A_j a - \sqrt{J} \phi\|^2$, which is positive semidefinite with eigenvalues $0, 2J$. The correct characterization of $c = 1$ is the single-operator Schur completion

$$\mathcal{A}(J) = D - J \Pi_n, \quad D = \begin{pmatrix} 0 & A_j^\dagger \\ A_j & 0 \end{pmatrix}, \quad D^2 = J \mathbf{1}, \quad (58)$$

with Π_n the projector onto the order-parameter channel: the first-order “Dirac” part D (the curl) supplies $\sqrt{J}$, and the rank-one term $-J \Pi_n = -A_j A_j^\dagger$ is the order-parameter self-energy built from the same A_j . The condition $c = 1$ is therefore that the order-parameter channel acquires its *entire* mass as $-A_j A_j^\dagger$, with no independent contribution. The Englert/torsion structure of Section 8.3 makes this natural — the off-diagonal and the diagonal are then both built from the same structure-constant tensor ϵ^{abc} — but turning “same tensor” into “same coefficient” requires the explicit linearized computation. The conjecture (21) is exactly the statement that the order-parameter self-energy is $-A_j A_j^\dagger$.

Remark 8.2 (Relation to the Breitenlohner–Freedman window). In an AdS_4 vacuum a negative mass-squared is not automatically an instability: scalars are stable down to the Breitenlohner–Freedman bound. The negative diagonal $-J$ should be read in that light — as a genuine relevant (order-parameter) direction in the dual three-dimensional sense, whose condensation is electroweak breaking. We do not pursue the $\text{AdS}_4/\text{CFT}_3$ dictionary here, but we note that it provides a controlled language for “the negative branch is the order parameter,” and it is the natural setting in which to attempt a proof of (21).

8.6 A conditional derivation of $c = 1$

We can do better than declaring $c = 1$ an ansatz. We give here a conditional derivation: a precise sufficient condition under which $c = 1$ is *forced*, a proof that the dangerous Weitzenböck curvature shift does not spoil it under that condition, and the explicit value of c when the condition fails. This reduces the open problem from “compute c ” to “establish one structural hypothesis,” and it makes the whole construction falsifiable in a sharp way.

On the unit S^3 there are two distinct second-order operators in play, and they are not equal. The scalar Laplacian is $\Delta_0 = 4J$ (Eq. (24)). The Hodge–de Rham Laplacian on transverse (co-exact) one-forms is, by the Weitzenböck–Lichnerowicz identity $\Delta_1 = \nabla^* \nabla + \text{Ric}$ with $\text{Ric} = 2g$ on the unit S^3 ,

$$\Delta_1|_{\text{co-exact}} = \Delta_0 + 1 = 4J + 1 = (2j + 1)^2, \quad (59)$$

shifted by one unit of curvature relative to the scalar. The de Vries off-diagonal, by contrast, is the *algebraic* Casimir root $A_j^\dagger A_j = J$ (Lemma 3.1), neither $4J$ nor $4J + 1$. The coefficient c is the ratio of the order-parameter diagonal mass to the off-diagonal squared. Writing the flux-induced quadratic terms as an off-diagonal coupling $\alpha \text{Re}\langle a, A_j \phi \rangle$ and a diagonal self-energy $-\beta \langle \phi, A_j^\dagger A_j \phi \rangle$, and normalizing the off-diagonal to $\sqrt{J}$, the coefficient is

$$c = \frac{\beta}{\alpha^2}, \quad (60)$$

and the question is whether it is forced to 1. The structure-constant contraction $\epsilon_{abc} \epsilon^{ab}_d = 2\delta_{cd}$ does reduce the flux-squared diagonal to the *same* Casimir $A_j^\dagger A_j = J$ that sets the off-diagonal — so parallelizability fixes the index structure — but it does *not* by itself force $\beta = \alpha^2$. There is moreover a real danger from curvature: if the off-diagonal is built from one operator and

the diagonal from the independently curvature-shifted other, the $+1$ of (59) survives in the ratio and c becomes j -dependent — which would destroy the relation, not merely shift it. The following proposition isolates the condition under which the danger is absent.

Proposition 8.3 (Conditional derivation of $c = 1$). *Suppose the order-parameter scalar ϕ in the spin- j sector (i) is a flat direction of the fluxless compactification, i.e. has vanishing bare and Freund–Rubin mass, and (ii) couples to the background flux through a single Stückelberg portal: its only flux coupling is the term $a^\dagger A_j \phi + \text{h.c.}$, with A_j the same first-order operator whose $A_j^\dagger A_j = J$ sets the off-diagonal. Then the reduced fluctuation block is the single-operator Schur completion (58), $\mathcal{A}(J) = [[0, A_j^\dagger], [A_j, -A_j A_j^\dagger]]$, so that the coefficient*

$$\boxed{c = 1} \tag{61}$$

is forced.

A clarification is essential here, and we were initially too sanguine about it. The Proposition fixes the *shape* of the block ($c = 1$); the de Vries *number* requires one further normalization, namely that the off-diagonal squared be the *bare* Casimir $A_j^\dagger A_j = J = j(j+1)$ of Lemma 3.1 — the algebraic operator $A_j = \sum_a T_a^{(j)} \otimes e_a$ — and not a curvature-shifted geometric Laplacian. This is where the Weitzenböck shift (59) genuinely bites: while c (the ratio diagonal/off-diagonal²) is invariant under a common rescaling of both entries, the weak *angle* depends on the *argument* of X_+ , so replacing the bare Casimir J by the shifted curl eigenvalue $4J+1$ changes the prediction. Numerically: with both entries the bare Casimir ($J = \frac{3}{4}, 2$) one gets $\sin^2 \theta_W = 0.2231$; with both entries the geometric curl square ($4J+1 = 4, 9$) one gets $\sin^2 \theta_W = 0.088$; the mismatched cases lie between (0.21–0.27). Only the bare-Casimir normalization gives the de Vries value. Hence the genuine content of the hypothesis is sharper than “ $c = 1$ ”: it is that the physical flux operator is the *algebraic Casimir* A_j (so off-diagonal² = J exactly), used for both entries (so $c = 1$). Equivalently, the single-portal flat modulus must couple through A_j itself rather than through a curvature-shifted geometric Laplacian.

Sketch. By (i) the diagonal of ϕ has no contribution other than the flux one. By (ii) the only flux term quadratic in the fields is $a^\dagger A_j \phi + \text{h.c.}$; integrating the order-parameter channel in its own equation of motion (or completing the rank-one self-energy generated by this single portal) gives the diagonal $-A_j A_j^\dagger$, the off-diagonal $A_j^\dagger$, and the gauge diagonal 0 (Axiom 1). This is (58), whose projection to the image gives $c = 1$. Because both non-trivial entries are built from the *same* A_j , multiplying the unit of $A_j^\dagger A_j$ by any common factor rescales off-diagonal and diagonal together and leaves $c = 1$; the curvature shift (59) is such a common factor. $\square$

The hypothesis is non-trivial, and its failure is concrete. Suppose instead the order-parameter diagonal is the *independent* transverse-vector Laplacian $4J+1$ while the off-diagonal is the scalar-side gradient $\sqrt{4J}$ — two different, separately curvature-shifted operators. Then $c = (4J+1)/(4J)$, which is j -dependent:

$$c_W = \frac{4(3/4) + 1}{4(3/4)} = \frac{4}{3}, \quad c_Z = \frac{4(2) + 1}{4(2)} = \frac{9}{8}. \tag{62}$$

Using $X_+(J; c) = \frac{1}{2}(-cJ + \sqrt{c^2 J^2 + 4J})$ this gives $X_+(\frac{3}{4}; \frac{4}{3}) = \frac{1}{2}$ and $X_+(2; \frac{9}{8}) = 0.6821$, hence

$$\sin^2 \theta_W|_{\text{independent}} = 1 - \frac{1/2}{0.6821} = 0.267, \tag{63}$$

far from the measured 0.2231. So the single-portal hypothesis of Proposition 8.3 is doing genuine, falsifiable work: it is the difference between 0.223 and 0.267, and it is exactly the statement that the order-parameter mass is $-A_j A_j^\dagger$ rather than an independent curvature-shifted Laplacian.

In sum, Proposition 8.3 establishes the reduction $c = 1 \Leftrightarrow$ single-operator Schur completion, the sufficient condition for it, the bare-Casimir normalization required for the de Vries number, and the explicit failure value $\sin^2 \theta_W = 0.267$. The structural realization is a flat geometric modulus — the shape λ (“Higgs = radion”) — lifted only by the flux that also mixes it with the gauge mode, in the manner of a Scherk–Schwarz / twisted-torus gauging where the structure constants f^{abc} appear linearly in the vector (Stückelberg) mass and quadratically in the scalar potential [16, 17], so the diagonal is $-A_j A_j^\dagger$ of the same A_j .

8.7 The single portal as gauge–Higgs unification

The single-portal condition of Proposition 8.3 is not an arbitrary hypothesis: it is the defining feature of an established framework, gauge–Higgs unification [18, 19, 20], in which the Higgs is the internal component of a higher-dimensional gauge field. We argue that the de Vries order parameter is of this type, which supplies both ingredients the de Vries number needs.

Split the higher-dimensional weak gauge field $\hat{A}_M$ into the four-dimensional gauge field A_μ and the internal components A_a on the weak $\mathbb{CP}^1 = \text{SU}(2)/\text{U}(1)$ (the Hopf base of $S^3 \cong \text{SU}(2)$); the A_a are the Higgs. Two properties are automatic, and they are conditions (i) and (ii) of Proposition 8.3.

1. *Flatness.* A mass term for A_a is forbidden by the higher-dimensional gauge invariance, so the Higgs is a Wilson-line (Hosotani) modulus — massless at tree level, lifted only by flux and loops. This is condition (i), vanishing bare mass.
2. *Single portal.* The Higgs reaches the four-dimensional gauge field only through the higher-dimensional field strength: the cross-component $F_{\mu a} = \partial_\mu A_a - D_a A_\mu$ supplies the gauge–Higgs mixing (the covariant-derivative/Stückelberg term), and the internal $F_{ab} = \partial_{[a} A_{b]} + [A_a, A_b]$ supplies the Hosotani potential. Both are built from the one gauge field through the structure constants; there is no independent Higgs coupling. This is condition (ii), the single portal.

On $\mathbb{CP}^1$ the mixing operator is the $\text{SU}(2)$ gauge-current zero mode on the line-bundle modes, $J_0^a|_{V_j} = T_a^{(j)}$, i.e. the algebraic operator $A_j = \sum_a T_a^{(j)} \otimes e_a$ with $A_j^\dagger A_j = \sum_a T_a T_a = J$. This is the gauge coupling $[A_\mu, \phi] = A_\mu^a \text{ad}(T_a)\phi$, the Lie-algebra action of the generators on the mode — *not* the metric curl $\star d$ on the cotangent bundle, which is the operator carrying the Weitzenböck term $\Delta_1 = 4J + 1$. One can see the bare Casimir appear explicitly in the standard S^2 reduction: with $A = \partial_\theta/R$ the Stückelberg pairing gives $A^\dagger A = \ell(\ell + 1)/R^2 = J/R^2$, the bare Casimir and not the shifted curl eigenvalue [18, 20]. *This is the gap that gauge–Higgs unification genuinely closes: the normalization is bare-Casimir, so the off-diagonal is $\sqrt{J}$ and not $\sqrt{4J + 1}$.*

It is important not to claim more. Gauge–Higgs unification supplies the operator and its normalization; it does *not*, by itself, hand one the indefinite Schur block. The explicit gauge–Higgs spectrum on S^2 is, per level, a would-be-Goldstone component A_θ that mixes with A_μ through $\sqrt{J}$, *together with a separate* physical scalar A_ϕ of mass J that does *not* mix — not the single indefinite block $\mathcal{A}(J) = [[0, A_j^\dagger], [A_j, -A_j A_j^\dagger]]$. Reorganizing the Goldstone mixing and the scalar mass into one Schur completion of a single operator is the single-portal identification of Proposition 8.3; it is the content of $c = 1$, and it is an assumption beyond what standard gauge–Higgs unification delivers. (On $S^3 = \text{SU}(2)$ the same caveat is visible in $F_{ab} = [\Phi_a, \Phi_b] - f_{ab}^c \Phi_c$, whose coset-geometry term $-f_{ab}^c \Phi_c$ adds curvature/torsion contributions to the internal Hessian beyond $\text{ad}(\Phi)^2$.) What gauge–Higgs unification therefore buys is the *normalization* (bare Casimir, the harder of the two structural worries) and a natural *home* for a flat, single-portal Higgs; it does not, on its own, force the Schur block.

Two honest qualifications bound what this buys, and they should be stated plainly. First, obtaining a weak angle from gauge–Higgs unification is itself classic — Manton and Fairlie did

so in 1979, finding $\sin^2 \theta_W = \frac{1}{4}$ for the $SU(3)/[SU(2) \times U(1)]$ model on S^2 [18, 19] — but they read it off the embedding *coupling ratio* g'/g . The de Vries value 0.2231 is a *different* observable: it is the on-shell *mass ratio* $1 - X_+(\frac{3}{4})/X_+(2)$ read from the poles of the Schur block at the two spins $j = \frac{1}{2}, 1$, not the coupling ratio. So 0.2231 is not the textbook gauge–Higgs $\frac{1}{4}$; identifying the physical W, Z masses with the Schur eigenvalues (rather than with the standard Higgs-mechanism masses) is an additional assumption of the de Vries reading, and to our knowledge this Schur/pole block has not appeared in the gauge–Higgs literature. Second, pure gauge–Higgs unification gives the Higgs *zero* tree-level mass; the negative diagonal $-J$ is generated by the Hosotani potential (radiatively, or by flux/curvature on the coset), and while the single-operator structure ties its sign and shape to the mixing, the statement that its magnitude is exactly $-A_j A_j^\dagger$ (and not a rescaled multiple) is the residual coefficient that the Hosotani computation must return.

What gauge–Higgs unification therefore contributes is genuine but bounded: it *derives* the bare-Casimir normalization — the harder of the worries, the one that fixes the off-diagonal at $\sqrt{J}$ and so the value 0.2231 rather than a curvature-shifted number — and it gives a natural home for a flat, single-portal Higgs. It does not by itself produce the indefinite Schur block. The open problem is thereby reduced to, and located in, three sharp residues: the single-portal/Schur identification (that the Goldstone mixing and the scalar mass are one operator’s completion), the Hosotani coefficient (that the magnitude is exactly $-A_j A_j^\dagger$), and the pole identification (that the W, Z masses are the Schur eigenvalues, the mass-ratio reading that gives 0.2231 rather than the coupling-ratio $\frac{1}{4}$). Each is a definite, answer-blind computation.

8.8 Self-consistency: our own skepticism applied to $c = 1$

We owe the reader a hard look at our own central assumption, by the same standard we apply to the classical M^{pqr} integer fit (Section 15). There we argue that reproducing the weak angle by tuning four integers is an over-determination, because the achievable values are dense around the target and a “distinguished” choice carries no selection information. The intellectually honest question is whether $c = 1$ is the same move with one knob instead of four.

In part, it is. The family $\sin^2 \theta_W(c)$ of Eq. (19) sweeps continuously through the measured value, taking 0.30 at $c = \frac{1}{2}$ and 0.21 at $c = 1.1$ (Table 1); the data do not by themselves prefer $c = 1$ over its neighbours any more than they prefer one integer tuple over another. Calling $c = 1$ “self-dual” or “the perfect-square point” is a distinguished-point argument, and distinguished-point arguments are exactly what we distrusted in the M^{pqr} case. We must therefore not claim that the empirical success of $c = 1$ is, by itself, evidence for the geometry; that would be the error we diagnosed, in a thinner disguise.

There are two honest differences, and they are differences of *degree*, not of kind. First, c is a *single* parameter with a structural distinguished value (the single-operator Schur completion, Eq. (58)), whereas the M^{pqr} fit has four integers and no structural reason for the winning tuple; a one-parameter family with a geometrically marked point is less free than a dense four-integer lattice, but it is not parameter-free. Second — and this is the decisive methodological point — $c = 1$ is *falsifiable by a computation we have specified*: the linearized Type IIA action on the stabilized K_6 either does or does not give the order-parameter channel the self-energy $-A_j A_j^\dagger$, and that computation does not know the experimental number. The M^{pqr} integer fit has no such pending decider; it is fit and then frozen. We therefore promote $c = 1$ above the integer fit not because it works numerically — it does, but so do many things — but because it is the unique value reducible to a single sharp, answer-blind calculation.

Until that calculation is done, the correct status of the weak angle is the conditional one: the geometry yields the family $\sin^2 \theta_W(c)$, the de Vries point $c = 1$ is a structurally distinguished but unproven ansatz, and the agreement with experiment is suggestive evidence *for the ansatz*, pending its independent test — not, yet, a parameter-free prediction of the geometry. We

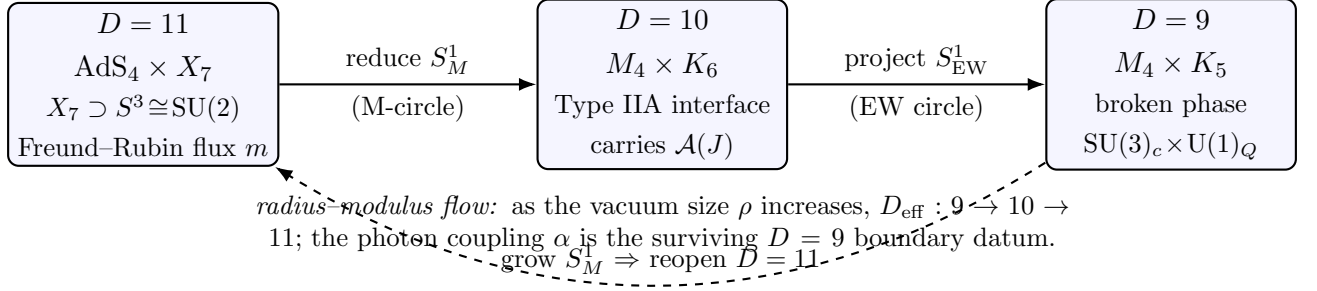


Figure 2: The dimensional ladder. Two distinct circle operations connect the eleven-dimensional Freund–Rubin parent to the nine-dimensional electroweak endpoint: the M-theory circle S_M^1 relates $D = 11$ and $D = 10$, while a different electroweak circle S_{EW}^1 relates $D = 10$ and $D = 9$. The electroweak operator $\mathcal{A}(J)$ lives on the $D = 10$ interface $K_6 \supset S^3$.

state this plainly because the alternative — quietly treating $c = 1$ as established — is the self-deception this paper was written to avoid.

9 The dimensional ladder: $D = 11 \rightarrow D = 10 \rightarrow D = 9$

We now assemble the geometric narrative connecting the eleven-dimensional parent to the nine-dimensional electroweak endpoint. The reason this ladder has three rungs, and not two, is a point that is easy to get wrong, and we take care with it.

9.1 Two circles, not one

The naive worry is a contradiction: how can one extra coordinate *decompactify* the theory to eleven dimensions (the M-theory circle growing) and at the same time *disappear* to leave a nine-dimensional electroweak endpoint? The resolution is that these are *two different circles* [15].

- **The M-theory circle S_M^1 .** Reducing eleven-dimensional supergravity on a circle gives ten-dimensional Type IIA string theory; the radius of S_M^1 is the IIA string coupling. Growing S_M^1 is the $D = 10 \rightarrow D = 11$ direction (strong coupling, M-theory). This circle is part of X_7 (it is, in the homogeneous models, the fibre of the $S^1 \rightarrow M^{pqr} \rightarrow \mathbb{CP}^2 \times \mathbb{CP}^1$ presentation).
- **The electroweak circle S_{EW}^1 .** A *different* compact direction — the one whose Wilson line / boundary projection implements electroweak symmetry breaking — shrinks (or is projected out by a T-dual/open-string boundary condition) in the $D = 10 \rightarrow D = 9$ direction, leaving the broken-phase endpoint with gauge group $\text{SU}(3)_c \times \text{U}(1)_Q$.

The two operations act on different coordinates, so there is no contradiction. The ladder is

$$\underbrace{\text{AdS}_4 \times X_7}_{D=11} \xrightarrow{S_M^1 \text{ reduce}} \underbrace{M_4 \times K_6}_{D=10} \xrightarrow{S_{\text{EW}}^1 \text{ project}} \underbrace{M_4 \times K_5}_{D=9}, \quad (64)$$

with $K_6 \supset S^3$ the interface that carries the electroweak operator and K_5 the broken-phase endpoint carrying colour and electric charge. Figure 2 summarizes the structure.

9.2 Probe energy and the $K_5/K_6/K_7$ ladder

The cleanest way to read the three “dimensions” is not as a change in the geometry but as a change in the *probe*. The internal geometry is fixed, with one overall scale; what a given experiment resolves depends on its energy. A probe of energy E resolves a compact direction of proper radius R — excites its Kaluza–Klein tower, whose gap is $1/R$ — precisely when

$$ER \gtrsim 1, \quad (65)$$

and is blind to it for $ER \ll 1$. The internal manifold is built from nested factors of increasing inverse radius (decreasing size),

$$K_5 \subset K_6 = K_5 \times S_{\text{EW}}^1 \subset K_7 = K_6 \times S_M^1, \quad (66)$$

the electroweak circle S_{EW}^1 and the M-theory circle S_M^1 being the two distinct circles of Section 6. As the probe energy climbs past the corresponding thresholds $1/R_{\text{EW}}$ and $1/R_M$, one additional internal dimension switches on at each step, so the number of resolved internal dimensions runs $5 \rightarrow 6 \rightarrow 7$:

IR probe	$E < 1/R_{\text{EW}} :$	K_5	$D = 9$	$\text{SU}(3)_c \times \text{U}(1)_Q$, only α survives;
EW-scale probe	$1/R_{\text{EW}} < E < 1/R_M :$	K_6	$D = 10$	electroweak S^3 resolved, $\mathcal{A}(J)$ lives here;
UV probe	$E > 1/R_M :$	K_7	$D = 11$	M-circle open, supergravity parent.

(67)

Thus the infrared experiment sees a five-dimensional internal space (the $D = 9$ endpoint, where only the electric coupling α is visible); an experiment at the electroweak scale sees six (the $D = 10$ interface carrying the de Vries operator); and a hypothetical ultraviolet probe sees seven (the $D = 11$ supergravity parent). The “ $9 \rightarrow 10 \rightarrow 11$ ” ladder is the *probe-energy* ordering, not a sequence of distinct theories.

We live in $D = 10$. The electroweak experiments that measure $\sin^2 \theta_W$, M_W , and M_Z operate at the middle rung: their probe energy resolves the electroweak S^3 but not the M-circle, so the world they see is ten-dimensional, $M_4 \times K_6$, the Type IIA interface on which $\mathcal{A}(J)$ is a fluctuation. The eleven-dimensional parent is reached only by a probe above $1/R_M$ (which we do not have), and the nine-dimensional endpoint only in the deep infrared after electroweak breaking. The middle rung is not one option among three; it is our home, and Section 14 shows that it is also the only rung on which the chiral fermions can live.

This is also the precise meaning of the construction’s having one physical scale. The geometry carries a single overall size (equivalently μ , the S^3 radius, or the flux scale m); all dimensionless content — the spins $j = 0, \frac{1}{2}, 1$, the flux normalization $c = 1$, the charge lattice — is topological or representation-theoretic and does not move with that scale. Different experiments do not change the theory; they change which rungs of the $K_5 \subset K_6 \subset K_7$ ladder lie below their resolution.

9.3 The role of Regge data

A word on where string Regge dynamics enters, since the relation is sometimes mistaken for a Regge trajectory. The de Vries positive branch is *bounded* ($X_+(J) < 1$), whereas a Regge trajectory $M^2 \sim J/\alpha'$ is unbounded. They are different objects: the de Vries block is the finite internal electroweak operator, while the genuine Regge tower is the string oscillator spectrum that sits *on top* of it,

$$M^2(n, j) = \frac{n - a_0}{\alpha'} + \mu^2 X_+(j(j+1)) + \Delta M_{\text{KK}}^2. \quad (68)$$

The string/Regge sector’s only role in the electroweak ratio is as a *selection principle*: it supplies the finite spin set $\{0, \frac{1}{2}, 1\}$ (Section 4). We are careful to use it *only* for that, and not for the

operator normalization: the level-two truncation fixes which representations appear, while the off-diagonal coefficient is fixed independently by the geometric curl ($A_j^\dagger A_j = J$, not the WZW weight $J/(k+2)$). Treating the finite set $\{0, \frac{1}{2}, 1\}$ as the electroweak content (an input, like the gauge group) and the curl as the operator is consistent; the two statements are logically separate. A direct identification of the de Vries branch with a Regge trajectory is incorrect and we do not make it.

9.4 Branches and brane dimension

There is, however, a clean place where the two branches of $\mathcal{A}(J)$ meet the spinning-object literature, and it is worth recording because it organizes the bounded and unbounded branches into one family. A spinning q -brane has the Regge scaling

$$M^2 \propto J^{2q/(q+1)}, \quad (69)$$

with the three viewpoints: $q = 0$ (a point particle / 0-brane) gives $M^2 \sim J^0 = \text{const}$, a *bounded* spectrum; $q = 1$ (the string) gives the familiar linear trajectory $M^2 \sim J$; and $q \rightarrow \infty$ (a space-filling brane) gives $M^2 \sim J^2$. The two branches of the de Vries operator sit at the first two of these endpoints. The positive branch is bounded, $X_+(J) \rightarrow 1$ as $J \rightarrow \infty$ — the $q = 0$, point-like behaviour appropriate to a finite internal (electroweak) operator. The negative branch grows linearly, $X_-(J) \rightarrow -(J+1)$ — the $q = 1$, string-like behaviour. And the *branch product*,

$$X_+(J) X_-(J) = -J, \quad (70)$$

is exactly the linear (string, $q = 1$) Regge invariant: it is the quantity that scales linearly in the Casimir, while neither branch alone does. The de Vries block thus packages a point-like (bounded) gauge branch and a string-like (linear) order-parameter branch into a single operator whose invariant is the linear trajectory. This is the limited sense in which Regge structure is present: not as the electroweak spectrum itself, but as the brane-dimension family whose $q = 0$ and $q = 1$ endpoints are the two de Vries branches.

10 The electromagnetic coupling at the $D = 9$ boundary

The electroweak ratio is free of continuous moduli (conditional on $c = 1$); the absolute electromagnetic coupling requires one more ingredient, the absolute size of the surviving $U(1)_Q$ generator. We collect the standard Kaluza–Klein normalization and the masslessness of the photon.

10.1 Couplings from circumferences

Weinberg’s rule [11] expresses a four-dimensional gauge coupling in terms of the proper length L of the orbit of the corresponding internal isometry,

$$g^2 = \frac{4\pi^2 \kappa^2}{L^2}, \quad \kappa^2 = 16\pi G, \quad (71)$$

with G the four-dimensional Newton constant. For the electroweak generators T_3 and Y , with orbit lengths L_2 and L_Y , the unbroken couplings are $g_2^2 = 4\pi^2 \kappa^2 / L_2^2$ and $g_Y^2 = 4\pi^2 \kappa^2 / L_Y^2$. The weak angle is $\tan^2 \theta_W = g_Y^2 / g_2^2 = L_2^2 / L_Y^2$, and after breaking the electromagnetic coupling is

$$\frac{1}{e^2} = \frac{1}{g_2^2} + \frac{1}{g_Y^2} = \frac{L_2^2 + L_Y^2}{4\pi^2 \kappa^2}, \quad \alpha_{\text{EM}} = \frac{16\pi^2 G}{L_2^2 + L_Y^2}. \quad (72)$$

The absence of a cross term $1/g_{2Y}$ in (72) — equivalently $L_Q^2 = L_2^2 + L_Y^2$ with no L_{2Y} — is not an assumption but follows from the structure of the gauge group: there is no Ad-invariant bilinear

form pairing the simple-SU(2) generator T_3 with the external $U(1)_Y$, so the gauge-kinetic matrix is diagonal in the (T_3, Y) basis. The diagonal α formula is therefore exact.

Equation (72) shows the division of labour cleanly: the *ratio* L_2/L_Y is the weak angle (which we obtain from $\mathcal{A}(J)$, not from the lengths), while the *absolute* lengths fix α . The de Vries expression (10) is the statement that the same geometry which fixes the ratio also fixes the combination $L_2^2 + L_Y^2$ that appears in α ; making this quantitative — deriving the absolute S^3 and $U(1)$ radii from the flux m — is part of the open problem of the absolute scale.

10.2 The photon stays massless

The surviving gauge generator is $Q = T_3 + Y$, and the photon’s masslessness is the statement that the order-parameter vacuum is Q -neutral. Take the electroweak doublet basis $\eta_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\eta_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with $T_a = \frac{1}{2}\sigma_a$, $Y = \frac{1}{2}$. The neutral component satisfies

$$T_3 \eta_0 = -\frac{1}{2} \eta_0, \quad Y \eta_0 = +\frac{1}{2} \eta_0, \quad \implies \quad Q \eta_0 = (T_3 + Y) \eta_0 = 0. \quad (73)$$

A vacuum $\Phi_0 = v_F \eta_0$ therefore leaves $U(1)_Q$ unbroken and the photon massless — the $j = 0$, $X = 0$ slot of $\mathcal{A}(J)$. The same neutral vector carries SU(2) Casimir $\sum_a \|T_a \eta_0\|^2 = \frac{3}{4} = J|_{j=1/2}$, tying the photon-protecting order parameter to the $j = \frac{1}{2}$ branch label.

In the geometry, η_0 is the Q -neutral section of the line bundle $\mathcal{O}(1)$ over the weak $\mathbb{CP}^1 = S^3/S_M^1$, and the surviving $U(1)_Q$ is the diagonal weak–hypercharge frame circle $P_Q = \ker \chi_N$, with $\chi_N(u_T, u_Y) = u_T^{-1} u_Y$ the neutral character. We will not need the details; the content is Eq. (73).

11 The negative branch as the Higgs / order parameter

We promised a reading of the negative branch. The interpretation is forced by the structure: the nn -entry of $\mathcal{A}(J)$ is a negative mass-squared, the defining property of an order-parameter (Higgs) direction.

11.1 The order parameter and v_{EW}

In the broken phase the standard kinetic term of the doublet $\Phi_0 = v_F \eta_0$ gives the gauge masses

$$M_W^2 = \frac{1}{4} g_2^2 v_{EW}^2, \quad M_Z^2 = \frac{1}{4} (g_2^2 + g_Y^2) v_{EW}^2, \quad M_\gamma^2 = 0, \quad (74)$$

with $v_{EW} = \sqrt{2} |v_F|$. The de Vries branch fixes the *ratio* M_W^2/M_Z^2 ; the order-parameter amplitude v_F fixes the overall scale. Comparing with Eq. (3), $\mu^2 X_+(2) = M_Z^2 = \frac{1}{4} (g_2^2 + g_Y^2) v_{EW}^2$, ties μ to v_{EW} and the couplings; numerically $\mu \simeq \sqrt{v M_Z/2} \approx 107 \text{ GeV}$, recovering the geometric-mean reading mentioned in (P2). The identification of the order-parameter modulus with the size modulus (“Higgs = radion”) is natural in this setting and would make v_{EW} a geometric quantity; we list this as an attractive but unproven option in Section 17.

11.2 The negative roots and the top/Higgs window

The magnitudes of the negative roots, with the same single scale, are

$$\mu \sqrt{|X_-(3/4)|} = 122.4 \text{ GeV}, \quad \mu \sqrt{|X_-(2)|} = 176.2 \text{ GeV}. \quad (75)$$

The proximity to $m_H = 125.25 \text{ GeV}$ and $m_t = 172.7 \text{ GeV}$ is suggestive, and it sharpens under the complex-pole reading of the relation (where the masses are real parts of complex poles). We stress that this is a *diagnostic*: we have not derived the identification of the negative $j = \frac{1}{2}$ root with the Higgs scalar or the negative $j = 1$ root with the top quark. But the fact that one

scale μ , fixed by M_Z , simultaneously places the positive branch on (M_W, M_Z) and the negative branch near (m_H, m_t) is part of what makes the relation hard to dismiss as a single accident. It is also exactly the pattern one expects if all four masses are the four roots of two copies of one quadratic — the deepest version of the de Vries claim, and the one most worth testing. It is suggestive that the negative-branch point $(m_t, m_H) \approx (176, 122)$ GeV lies in the same near-critical region of the Standard Model (m_t, m_H) plane mapped out by the vacuum-stability analyses [22], an entirely independent calculation that knows nothing of $\mathcal{A}(J)$; we record this as a coincidence worth watching, not as evidence.

11.3 The Higgs mass as a curvature

The Higgs mass is, in this language, the radial curvature of the order-parameter potential at its minimum,

$$M_H^2 = \frac{1}{Z_H} \left. \frac{\partial^2 V_{\text{eff}}}{\partial h^2} \right|_*, \quad (76)$$

with Z_H the kinetic normalization of the radial mode h . Deriving V_{eff} from the flux compactification — and hence M_H and v from the same data that give M_W/M_Z — is the scalar-sector analogue of the flux-coefficient problem (21). We do not solve it; we note that the negative-branch value 122.4 GeV is the target.

11.4 Higgs as radion

The most economical scenario identifies the order-parameter modulus with the size modulus of the internal geometry — the Higgs field is the radion of the electroweak S^3 . This is attractive for three reasons. First, it removes a field: the same modulus that stabilizes the compactification and the field that breaks electroweak symmetry are one and the same, so no separate Higgs sector need be postulated. Second, it explains the “geometric mean” value of the mixing scale: if v_{EW} and M_Z are both functions of the single radion, the natural electroweak scale is their geometric mean $\mu \sim \sqrt{v M_Z/2} \approx 107$ GeV, exactly the value extracted in (P2). Third, it makes the Higgs couplings predictions rather than inputs: a radion Higgs has definite, generically non-standard couplings $g_{hWW}, g_{hZZ}, g_{hff}$, providing a falsifiable coupling-pattern test. Precedents for “Higgs as a geometric modulus” in extra-dimensional models are well known, and the present setting — a relevant order-parameter direction of the de Vries block whose condensation is breaking — is a natural home for the idea. We list it as the preferred but unproven scenario for the absolute scale.

11.5 The two-sheet (finite-distance) reading

There is a complementary, useful language for the order parameter, due in spirit to Connes’s noncommutative-geometry picture of the Standard Model. The neutral section $\eta_0 \in H^0(\mathbb{CP}^1, \mathcal{O}(1))$ that protects the photon (Section 10) can be read as a connection in a *finite* (two-point) extra direction — two “sheets” separated by a distance set by the inverse order-parameter amplitude,

$$\ell_H = \frac{1}{\|\Phi_0\|} = \frac{\sqrt{2}}{v_{\text{EW}}}. \quad (77)$$

In this reading the electroweak scale is literally a distance — the gap between the sheets — and the Higgs mass is the curvature of the finite-direction potential.

This reading does more than provide a picture: it is a *limit of the construction that avoids the Kaluza–Klein tower*. The objection to any S^3 -based electroweak sector is the infinite tower $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ (Section 6.5); one wants only the finite set $\{0, \frac{1}{2}, 1\}$. The Connes limit supplies exactly that. As the weak $\mathbb{CP}^1$ (equivalently the Hopf S^3) is taken to a degenerate, finite-distance configuration, the internal geometry collapses to a *finite spectral triple* — a finite-dimensional

algebra with a finite Dirac operator — whose spectrum is a finite set rather than a tower. In that limit the off-diagonal curl $\star d$ becomes a finite matrix (the finite-direction Dirac operator), and the de Vries block $\mathcal{A}(J)$ is its fluctuation Hessian, with the photon, W , and Z the entire spectrum and no heavy partners at all. The Connes (finite spectral triple) limit is therefore the tower-free face of the same construction: where the geometric description (Section 5) realizes $\{0, \frac{1}{2}, 1\}$ as the lowest rungs of a tower, the noncommutative limit realizes them as the *complete* spectrum of a finite internal space. The two are continuously connected by shrinking the weak circle, and the electroweak block is the common content. Making the limit precise — and showing the de Vries block is the spectral-action fluctuation Hessian of the finite triple — is open; we record it because it removes the tower objection at the cost of replacing the smooth S^3 by its finite (noncommutative) degeneration.

11.6 Rigidity: why the model cannot reach $M_W = M_Z$

A natural objection to any single-operator account of W and Z is that it might collapse the two into one. The de Vries block answers this cleanly: $M_W = M_Z$ would require $X_+(3/4) = X_+(2)$, i.e. $\frac{3}{4} = 2$, which is false; the two masses are *rigidly* separated because $j = \frac{1}{2}$ and $j = 1$ are different representations. Equivalently, in the g -language, $M_W = M_Z$ needs $g_Y = 0$, and the de Vries structure forbids it because the hypercharge mixing is locked to the $j = 1$ Casimir. This rigidity is a feature, not a bug: it is the statement that the weak angle is a fixed number, and it is the reason the relation is falsifiable. The separation $M_W < M_Z$ (rather than the reverse) follows from the monotonicity of $X_+(J)$ and the ordering $j_W = \frac{1}{2} < j_Z = 1$ — the lighter gauge boson sits on the lower internal representation.

12 Relation to other pictures of electroweak symmetry breaking

The branch operator $\mathcal{A}(J)$ is an unusual way to organize electroweak mass generation, and it is worth situating it among the standard pictures, both to clarify what is new and to inherit their constraints.

12.1 Custodial symmetry and $\rho = 1$

Any account of W and Z masses must respect the custodial $SU(2)$ that protects the tree-level relation $\rho \equiv M_W^2/(M_Z^2 \cos^2 \theta_W) = 1$. In the de Vries assignment this is automatic: with $M_W^2 = \mu^2 X_+(3/4)$, $M_Z^2 = \mu^2 X_+(2)$, and $\cos^2 \theta_W = X_+(3/4)/X_+(2)$ by construction (4), one has

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = \frac{X_+(3/4)}{X_+(2) \cdot X_+(3/4)/X_+(2)} = 1 \quad (78)$$

identically. The relation $\rho = 1$ is thus not an extra input but a consistency property of defining the weak angle through the same operator that gives the masses. This is reassuring: a generic two-channel mass mixing would violate $\rho = 1$ and be excluded by precision data at the per-mille level; the de Vries structure passes because $\cos^2 \theta_W$ is *defined* as the mass ratio.

12.2 Higgsless and boundary-condition breaking

The resolvent form of $\mathcal{A}(J)$ (Appendix A), $X = J/(J + X)$, is the hallmark of *boundary-condition* electroweak breaking, as in five-dimensional Higgsless and gauge-Higgs unification models: there, gauge-boson masses are the zeros of a transcendental function obtained by imposing boundary conditions at the ends of an interval, and the lightest modes solve a one-pole relation of exactly this type. The de Vries block is the minimal, two-channel truncation of such a boundary problem, with the “interval” being the radial direction of the electroweak S^3 and the boundary condition being the order-parameter vacuum. This connection is encouraging

on two counts: boundary-condition models are known to give $\rho = 1$ and acceptable precision-electroweak fits when custodial symmetry is built in, and they realize the off-diagonal mixing as a genuine geometric (Kaluza–Klein) effect rather than a Yukawa coupling. The de Vries relation can be read as the statement that the electroweak interval is the round S^3 , whose boundary spectrum is fixed by $SU(2)$ representation theory.

12.3 Magnetic-dual and partially composite readings

The appearance of the negative branch as an order parameter invites a magnetic-dual reading: $j = \frac{1}{2}$ may label an electric (or dual) variable whose W is a magnetic/IR pole, with W and Z partially composite. In such Seiberg-dual or partially composite scenarios the two-vector content (W as a magnetic pole, the elementary gauge field as the electric one) is the resolution of the apparent puzzle that one $SU(2)$ representation must furnish two physical vectors. We do not commit to this reading — it requires a strongly coupled dual we have not constructed — but it is the natural language for “the de Vries block maps one internal representation into a gauge pole and an order-parameter pole,” and it connects the construction to the well-developed literature on composite and dual electroweak sectors. The branch product $X_+ X_- = -J$ is the dual-pair invariant in this language.

12.4 The seesaw analogy

Finally, the structure $X^2 + JX - J = 0$ is formally a seesaw: a light state ($X_+ \rightarrow \sqrt{J}$ for small J , $X_+ \rightarrow 1$ for large J) and a heavy state ($X_- \rightarrow -(J+1)$ for large J) mixing through an off-diagonal $\sqrt{J}$, with the product of masses fixed to $-J$. The familiar neutrino seesaw has a *constant* heavy scale; here the “heavy” scale is the J -dependent order-parameter mass, and the off-diagonal is the J -dependent curl mixing, so the two are locked. In terms of the physical mass-squares $M_\pm^2 = \mu^2 X_\pm(J)$ (with $M_-^2 < 0$ the order-parameter mass) the reciprocal relation reads

$$\frac{1}{M_+^2} + \frac{1}{M_-^2} = \frac{1}{\mu^2} \frac{X_+ + X_-}{X_+ X_-} = \frac{1}{\mu^2} \frac{-J}{-J} = \frac{1}{\mu^2}, \quad (79)$$

the inverted (parallel-resistance) form of the relation, and another way to see that the two branches are the two roots of one structure rather than independent objects.

13 Global form and the charge lattice

A brief, standard, and genuinely derivable piece: the global form of the gauge group along the ladder. The unbroken Standard Model gauge group is

$$G_{\text{SM}} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6}, \quad y = 6Y, \quad (80)$$

the $\mathbb{Z}_6$ being generated by $\zeta_6 = (\omega_3, -1, e^{i\pi/3})$ with $\omega_3 = e^{2\pi i/3}$. On a state of colour triality t , weak weight $T_3 = m$, and charge $Q = m + Y$,

$$\zeta_6 : \omega_3^t e^{2\pi i m} e^{i\pi y/3} = \omega_3^t e^{2\pi i Q} = \omega_3^t e^{2\pi i q_3/3}, \quad q_3 = 3Q, \quad (81)$$

so after electroweak breaking the surviving identification is $\zeta_6 \mapsto \zeta_3 = (\omega_3, e^{2\pi i/3})$ with $\zeta_3^3 = 1$, and the $D = 9$ endpoint group is

$$G_{D9} = \frac{SU(3)_c \times U(1)_Q}{\mathbb{Z}_3}. \quad (82)$$

This $\mathbb{Z}_6 \rightarrow \mathbb{Z}_3$ descent is the global-form content of the ladder; in the geometry it is carried by the diagonal line bundle $c_1(L_N) = c_1(L_Y) - c_1(L_{T_3})$ of the neutral character χ_N . It is the one part of the “ M^{pqr} ” package that survives our critique below intact, because it is topology, not a fit.

14 Fermions, chirality, and why we live in $D = 10$

The construction so far concerns the electroweak *bosons*: the gauge fields from isometries and the mass operator $\mathcal{A}(J)$ from flux. The fermions force a sharpening of the whole picture, and that sharpening is the deepest structural statement of the paper: *we live in $D = 10$* . The eleven-dimensional supergravity is only the ultraviolet parent; the nine-dimensional endpoint is only the infrared limit; the physics we measure — including the chiral fermions — lives at the ten-dimensional, Type IIA string interface K_6 . The chirality problem is what makes this unavoidable.

14.1 The chirality obstruction in $D = 11$

The fermions of the Standard Model are chiral: left- and right-handed components transform differently under $SU(2)_L \times U(1)_Y$. Witten’s theorem [5] is that one *cannot* obtain a chiral four-dimensional spectrum from the Kaluza–Klein reduction of eleven-dimensional supergravity on a *smooth* compact seven-manifold: the four-dimensional massless fermions are the zero modes of the internal Dirac operator $\not{D}_K$, their net chirality is the index $\text{ind } \not{D}_K$, and on a smooth boundaryless homogeneous space of the M^{pqr} type this index does not deliver chiral generations — left and right modes pair up. This is the single most serious reason the classical eleven-dimensional Kaluza–Klein program of the early 1980s did not become the Standard Model. It is an obstruction of the *smooth $D = 11$ reduction*, and it is sharp.

14.2 Why the boson result is unaffected

The obstruction does not touch the electroweak boson result, and it is worth saying why before resolving it. The mass operator $\mathcal{A}(J)$ lives in the *bosonic* fluctuation spectrum — the metric, the three-form, and their flux mixing on the S^3 factor — governed by the Laplacian and the curl, not by the Dirac index. The weak mixing angle is a statement about how the gauge bosons distribute over internal $SU(2)$ representations, insensitive to the fermion sector, exactly as the gauge group itself is an isometry statement independent of the fermions. The boson prediction $\sin^2 \theta_W = 0.2231$ therefore stands on its own.

14.3 We live in $D = 10$: chirality from branes

The resolution of the chirality obstruction is not to deform the smooth eleven-dimensional reduction but to recognize that the theory we actually inhabit is *ten*-dimensional. Reducing eleven-dimensional supergravity on the M-theory circle S^1_M gives Type IIA string theory in $D = 10$, and Type IIA has an ingredient that smooth supergravity lacks: *D-branes*. Chiral fermions are then routine. They arise as open-string zero modes at the intersections of D-branes, or equivalently from D-branes wrapping cycles at ADE singularities, with the net chirality given by a topological intersection number rather than by the (vanishing) smooth Dirac index. This is the standard, and successful, string route to the chiral Standard Model: intersecting-brane and brane-at-singularity constructions in Type II string theory. In our setting the relevant branes are the D-branes wrapping the electroweak $K_6 = (S^3 \cong SU(2) \text{ sector})$ and the colour/hypercharge cycles, with chiral matter localized at their intersections.

The eleven-dimensional and ten-dimensional pictures are the two faces of one structure, related by the M-circle of Section 9.2. The chiral matter that appears in $D = 10$ at brane intersections lifts, in the $D = 11$ M-theory limit, to the Acharya–Witten mechanism [7]: chiral fermions at codimension-seven conical singularities of a G_2 -holonomy space, where D6-branes become loci of ADE singularity. The two descriptions agree, but the *physical, weakly coupled* description — the one in which both the de Vries boson operator and the chiral fermions are simultaneously visible and computable — is the ten-dimensional one. This is the precise content of “we live in $D = 10$ ”:

- $D = 11$ (the smooth M^{pqr} parent, K_7) is the ultraviolet completion; it gives the isometries (the gauge group) and the Freund–Rubin flux, but *cannot*, while smooth, give chiral fermions;
- $D = 10$ (Type IIA on $M_4 \times K_6$, $K_6 \supset S^3$) is where we live; it carries the de Vries operator $\mathcal{A}(J)$ and, through its D-branes, the chiral fermions, the Yukawa couplings, and the flavour structure;
- $D = 9$ (K_5) is the infrared endpoint after electroweak breaking, where only $SU(3)_c \times U(1)_Q$ and the coupling α survive.

The de Vries relation is thus a *ten-dimensional* statement: the electroweak operator is a fluctuation of the Type IIA interface, and the same ten-dimensional frame is the one in which chirality is solved. We do not construct the explicit brane configuration that yields three generations, the Yukawas, and the CKM matrix; that is the frontier. What the chirality problem establishes is the *location* of the physics — $D = 10$ — and that location is exactly where the de Vries operator was found.

14.4 Chirality across the ladder: why only $D = 10$ is delicate

The three rungs stand in a clarifying relation to chirality, and it is the relation that fixes $D = 10$ as the home of the fermions just as it is the home of $\mathcal{A}(J)$.

- **$D = 11$: chirality fails.** The smooth eleven-dimensional reduction cannot produce chiral fermions (Witten’s no-go). But we do not live there: $D = 11$ is the ultraviolet M-theory parent, reached only above the M-circle threshold.
- **$D = 10$: chirality is realized, and is the whole point.** The electroweak group $SU(2)_L \times U(1)_Y$ is *chiral* — left- and right-handed fermions carry different charges — and this chiral structure is an electroweak, i.e. ten-dimensional, phenomenon. It is realized by the D-branes of the Type IIA interface, whose open-string intersection modes are chiral. Chirality is delicate here, and here it is solved.
- **$D = 9$: chirality does not even arise.** After electroweak breaking the surviving group is $SU(3)_c \times U(1)_Q$ — QCD plus electromagnetism — which is *vector-like*: left- and right-handed fields carry identical colour and electric charges, so the fermions are ordinary Dirac (vector) fermions and no chiral index need be non-zero. The infrared endpoint therefore has *no* chirality problem to fail; there is nothing to obstruct.

The pattern is that chirality is born and resolved together, at the middle rung. It is absent in the infrared (the residual group is vector), impossible in the smooth ultraviolet (the no-go), and exactly realized in between, in the ten-dimensional electroweak world where the weak interaction is chiral and the de Vries operator lives. That the same rung carries both the parity-violating fermions and the boson mass relation is, we think, the strongest structural reason to read the de Vries relation as a statement about $D = 10$.

Each ingredient of this argument is standard: the impossibility of chiral fermions from smooth eleven-dimensional reduction is Witten’s theorem [5]; the appearance of chiral matter from D-branes is the basis of string phenomenology; and the vector-like character of QCD plus electromagnetism is elementary. What is perhaps not standard is the synthesis — that these three facts *together* pin the physical, chirality-bearing rung to $D = 10$, with both neighbours chirality-trivial for opposite reasons — and that this is the same rung on which the electroweak boson mass ratio is geometric. The argument uses nothing beyond textbook results, which is a point in its favour: a robust conclusion should follow from robust inputs.

14.5 A remark on the global form and the lattice

The charge-lattice and global-form data of Section 13 are the bridge between the two sectors. The $\mathbb{Z}_6$ identification, the hypercharge normalization $y = 6Y$, and the descent to $\mathbb{Z}_3$ are exactly the data that a consistent chiral spectrum must respect, and they are carried by the same line bundles over the weak $\mathbb{CP}^1$ that furnish the boson slots $j = 0, \frac{1}{2}, 1$. That the boson construction already fixes the global form to be the one compatible with Standard Model hypercharges is a nontrivial consistency check, and a hint that the singular completion, when built, will find the charge lattice waiting for it.

15 The M^{pqr} coupling-ratio program and the over-determination of the weak angle

The classical Kaluza–Klein literature suggests a different, tempting route to the weak angle — directly from the M^{pqr} orbit lengths — and it is important to establish quantitatively why that route does not predict the angle, and why we obtain it instead from $\mathcal{A}(J)$.

15.1 The Bailin–Love coupling ratios

Weinberg [11] and Bailin and Love [12, 13] computed the gauge-coupling ratios on M^{pqr} via the orbit-length rule (41). The weak-to-hypercharge ratio is a definite function of the integer labels (p, q, r, n) and the Einstein shape parameter β fixed by q/p ,

$$4\beta^3 - 6\beta^2 + \left(\frac{9}{4} + \frac{q^2}{p^2}\right)\beta - \frac{1}{2}\frac{q^2}{p^2} = 0 \quad (0 < \beta < \tfrac{1}{2}), \quad \frac{g_Y^2}{g_2^2} = \frac{3}{2} \frac{p^2 n^2}{r^2} \frac{(3 - 4\beta)^2 (1 + \beta)}{1 - 2\beta} \equiv F\left(\frac{q}{p}\right) \frac{p^2 n^2}{r^2}. \quad (83)$$

The structure is a continuous function $F(q/p)$ of the shape, multiplied by the rational $p^2 n^2 / r^2$ from the integer winding and charge-normalization data.

15.2 The integer fit

Scanning (83) for the de Vries value, the labels $(p, q, r, n) = (12, 14, 91, 1)$ give $\beta = 0.328481702\dots$ and

$$\frac{g_Y^2}{g_2^2} \Big|_{(12,14,91,1)} = 0.2871699126, \quad \tan^2 \theta_{\text{dV}} = 0.2871691363, \quad (84)$$

a match to a relative 2.7×10^{-6} . Taken at face value this looks like a six-digit prediction. It is not, for the reason we now make quantitative.

15.3 Over-determination: the family is dense around the target

The map $(p, q, r, n) \mapsto F(q/p) p^2 n^2 / r^2$ has four integer inputs constrained by one measurement, $\sin^2 \theta_W$, itself known only to $\sim 10^{-3}$. For any fixed shape q/p the residual factor $(np/r)^2$ ranges over the *squares of the positive rationals*, which are dense in $\mathbb{R}_{>0}$; layering the continuous variation of $F(q/p)$ over Farey fractions q/p makes the achievable set dense around the target. A direct scan over modest integers ($p, q \leq 30$, $r \leq 300$, $n \leq 3$) bears this out: there are 138 distinct geometries reproducing the measured weak angle to within experimental error (10^{-3}), 12 matching to 10^{-4} , and 2 to 10^{-5} . A representative sample of *independent* geometries (distinct β , not rescalings) is given in Table 2; every row reproduces $\tan^2 \theta_{\text{dV}}$ better than experiment can resolve.

A datum of relative precision $\delta \sim 10^{-3}$ can carry at most $\sim \ln(\text{range}/\delta)$ of selection information; the pre-image of the δ -ball around the target contains $O(10^2)$ integer tuples already in

(p, q, r, n)	q/p	β	g_Y^2/g_2^2	rel. dev.
(12, 14, 91, 1)	1.1667	0.328482	0.2871699	2.7×10^{-6}
(7, 26, 293, 2)	3.7143	0.490419	0.2871666	8.8×10^{-6}
(11, 11, 159, 2)	1.0000	0.250000	0.2871722	1.1×10^{-5}
(14, 21, 125, 1)	1.5000	0.418130	0.2871736	1.6×10^{-5}
(22, 25, 165, 1)	1.1364	0.315495	0.2871638	1.9×10^{-5}
(11, 5, 76, 1)	0.4545	0.047357	0.2871758	2.3×10^{-5}

Table 2: Independent M^{pqr} geometries reproducing the de Vries weak angle. Each has a distinct Einstein shape β ; all match the measured value to better than experimental precision. The advertised (12, 14, 91, 1) is one of many, and its 2.7×10^{-6} deviation is finer than the data warrant. The match is a property of the density of the family, not a selection of the manifold.

the smallest scan and diverges as the cutoffs grow. The likelihood that the Standard Model “selects” (12, 14, 91, 1) rather than one of its competitors is therefore $\lesssim 1/138$: the integer match is a fit, not a prediction.

15.4 Consequence for the present construction

Accordingly the weak angle in this paper is *not* taken from (83). It is taken from the parameter-free representation theory of $\mathcal{A}(J)$ (Corollary 3.3), whose only inputs are the spins $j = \frac{1}{2}, 1$ and the flux normalization $c = 1$ — a single statement (21), not a four-integer search. The M^{pqr} geometry enters only as the $D = 11$ grow-limit of the minimal K_6 (Section 7), through its isometries (the gauge group), its topology, and its charge lattice (Section 13); its orbit lengths fix the absolute scale of α through (42) but not the angle. The methodological discussion of why integer fits of this kind are to be avoided is collected in Appendix I.

16 Predictions and falsifiability

A physical theory earns its keep by making falsifiable statements. Here are the ones that follow from the construction.

(F1) The weak angle is a fixed number. $\sin^2 \theta_W = 0.22310$, on-shell, with no scale or scheme freedom in the ratio. This already agrees with experiment to 2×10^{-4} . Any future shift of the measured on-shell value by more than $\sim 10^{-3}$ would falsify the $c = 1$ form (cf. Table 1).

(F2) A neutral spin-zero state near 96.5 GeV. This prediction is conditional on the truncation mechanism. In the *physical-tower* (energetic) reading, the $j = \frac{3}{2}$ rung is a heavy but real state, a Q -neutral scalar at $M_{3/2} = \mu \sqrt{X_+(15/4)} = 96.5 \text{ GeV}$ — intriguingly close to the persistent $\sim 95\text{--}96 \text{ GeV}$ excesses in light-Higgs searches. In the *finite* (level-two / Connes, Section 11.5) reading the spectrum is exactly $\{0, \frac{1}{2}, 1\}$ and there is no such state. The two readings are thus distinguished by the presence or absence of the 96.5 GeV scalar: its discovery would select the physical tower, its exclusion the finite internal geometry.

(F3) The two-spin tower. The relation generalizes to any pair of internal spins,

$$\sin^2 \theta(j_a, j_b) = 1 - \frac{X_+(J_a)}{X_+(J_b)}, \quad J_i = j_i(j_i + 1), \quad (85)$$

with the electroweak case $(j_a, j_b) = (\frac{1}{2}, 1)$. Sample values: $\sin^2 \theta(\frac{1}{2}, \frac{3}{2}) = 0.307$, $\sin^2 \theta(1, \frac{3}{2}) = 0.108$, $\sin^2 \theta(\frac{1}{2}, 2) = 0.349$. If any further gauge or scalar splitting in an extended sector follows this family, it is a direct signature of the same operator.

(F4) The negative branch. The model predicts that the Higgs and top masses are the negative-branch partners of the W and Z , $m_H \approx \mu \sqrt{|X_-(3/4)|} = 122 \text{ GeV}$ and $m_t \approx \mu \sqrt{|X_-(2)|} = 176 \text{ GeV}$. The pattern is testable as precision on m_t , m_H , and the pole-mass definitions improves; the relevant comparison is to the pole (not $\overline{\text{MS}}$) masses and to the vacuum-stability boundary.

(F5) No tuned weak-angle integers. A specific *negative* prediction: the correct theory does *not* select $\sin^2 \theta_W$ by an M^{pqr} -type integer fit. If a first-principles derivation of the labels $(p, q, r, n) = (12, 14, 91, 1)$ from a dynamical principle were ever produced, it would be in tension with the present claim that the angle comes parameter-free from $\mathcal{A}(J)$ with $c = 1$. The two routes predict the *same* number but have different theoretical content, and they are distinguishable by whether the absolute coupling α and the $j = \frac{3}{2}$ state come out right.

17 Open problems

We list the genuine gaps, in decreasing order of importance, so that the status of the program is unambiguous.

- 1. The flux coefficient $c = 1$, and three residues.** Proposition 8.3 reduces “compute c ” to the single-portal condition, and Section 8.7 identifies that condition with gauge–Higgs unification (the Higgs as the internal weak gauge component). Gauge–Higgs unification *derives* the bare-Casimir normalization — because the portal is the algebraic gauge current T_a (Casimir J), not the curvature-shifted curl — and supplies a natural home for a flat, single-portal Higgs; it does *not*, on its own, produce the indefinite Schur block (the standard S^2 gauge–Higgs spectrum is a Goldstone with $\sqrt{J}$ mixing plus a separate scalar of mass J). Three sharp residues remain: (a) the single-portal/Schur identification (that the Goldstone mixing and the scalar mass are one operator’s completion); (b) the Hosotani coefficient (that the magnitude of the negative diagonal is exactly $-A_j A_j^\dagger$, from the flux/loop potential on the weak $\mathbb{CP}^1/S^3$); and (c) the pole identification (that the W, Z masses are the Schur eigenvalues, the mass-ratio reading giving 0.2231, rather than the embedding coupling ratio giving the textbook $\frac{1}{4}$). Each is a definite, answer-blind computation, and together they are the most important open problem.
- 2. The absolute scale.** The relation fixes ratios; one dimensionful input (μ , or the S^3 radius, or v_{EW}) remains. Tying μ to the flux m and ultimately to the Planck scale — and in particular deriving the absolute orbit lengths L_2, L_Y that fix α through (72) — is open. We recall the cautionary result of Ezawa and Koh [21], who found that the raw $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ couplings of the unbroken eleven-dimensional compactification come out far from the observed strengths; this is consistent with our position that the absolute scale (and the running to it) is unresolved, even though the *ratio* that gives the weak angle is not. The “Higgs = radion” identification, if correct, would make v_{EW} geometric.
- 3. The renormalization scheme of $\sin^2 \theta_W$.** The predicted 0.22310 is the *on-shell* angle $1 - M_W^2/M_Z^2$, and it should be compared only to the on-shell value 0.22305(20), not to the $\overline{\text{MS}}$ coupling angle $\hat{s}_Z^2 \simeq 0.23122$. A geometric construction of *couplings* would more naturally yield an $\overline{\text{MS}}$ -like quantity; that ours lands on the on-shell mass ratio reflects that $\mathcal{A}(J)$ is a mass operator, not a coupling operator. Making the scheme assignment precise — and likewise placing the $1/\alpha = 135.3$ of (10) on a definite scale between $\alpha(0)^{-1} = 137.036$ and $\alpha(M_Z)^{-1} \approx 128$ — is open.

4. **The vacuum: cosmological constant and AdS₄ versus Minkowski.** The Freund–Rubin limit is an AdS₄ vacuum, whereas the observed four-dimensional spacetime is nearly Minkowski. Uplifting to a small or zero cosmological constant, and confirming that the electroweak block survives the uplift, is an unaddressed standard problem; we have used AdS₄ only as the controlled setting in which the flux spectrum is known.
5. **Moduli stabilization.** The construction carries at least the overall scale ρ and the shape modulus λ . We have argued that λ plays the role of the order parameter (Higgs/radion), but we have not exhibited a stabilizing potential that fixes ρ and gives λ the tachyonic curvature $-J$ with the right coefficient; this is the same computation as the $c = 1$ problem viewed in the moduli potential.
6. **Spin structure for fermions.** $\mathbb{CP}^2$ is not spin; it is spin^c . Any fermion construction on K_6 must use the spin^c structure (a $U(1)$ gauge bundle tensored with the spinors), which is in fact natural here because the hypercharge line bundle supplies exactly such a $U(1)$. We note this as a constraint the (deferred) fermion sector must respect, not as a difficulty for the boson result.
7. **Chiral fermions and generations.** We have computed the electroweak *boson* sector. Chiral fermions, three generations, the Yukawa/CKM structure, and the strong sector are untouched and are the hard core of any realistic Kaluza–Klein program; the classical no-go theorems on chirality from smooth compactification [5] apply and likely force a singular (G_2/ADE) refinement of X_7 .
8. **The branch/orbit normalization subtlety.** The $j = \frac{1}{2}$ Casimir is $\frac{3}{4}$, while the literal charged-Higgs orbit norm $\|T_+ H_0\|^2 / \|H_0\|^2$ is 1. Both are correct statements about the same doublet; the de Vries spectrum uses the Casimir $\frac{3}{4}$ as the branch label, and the map to the physical W -pole normalization is a real conceptual point that deserves a cleaner treatment than we have given.
9. **A rigorous X_7 .** We kept X_7 schematic. Specifying a concrete Freund–Rubin seven-manifold that (a) contains the electroweak $S^3 \cong \text{SU}(2)$ at the self-dual point, (b) carries colour and hypercharge with the right charge lattice, and (c) admits the chiral refinement of point (4), is the geometric completion of the program.

18 Conclusions

We began with an arithmetic fact: the on-shell weak mixing angle and the W/Z mass ratio are reproduced, with no free parameter, by the positive eigenvalue of a 2×2 matrix $\mathcal{A}(J)$ evaluated at the $\text{SU}(2)$ Casimirs of $j = \frac{1}{2}$ and $j = 1$. We have argued that this fact is the shadow of a Kaluza–Klein theory of a very classical kind. The pieces fit together with little slack:

- the diagonal Casimir $J = j(j + 1)$ is one quarter of the scalar Laplacian on an internal $S^3 \cong \text{SU}(2)$, and the half-integer $j = \frac{1}{2}$ is the Peter–Weyl fingerprint of the group manifold rather than its $\text{SO}(3)$ quotient;
- the off-diagonal $\sqrt{J}$ is the matrix element of the curl $\star d$, a first-order, unit-normalized, self-adjoint operator whose square is the Laplacian — which is also why the electroweak masses are a bounded doublet rather than an unbounded tower;
- the negative diagonal $-J$ is the form-degree mixing induced by the $D = 10$ Type IIA background flux on K_6 , and the coefficient $c = 1$ is *conditionally derived* (Proposition 8.3): it is forced when the order parameter is a single-portal flat modulus, which Section 8.7 identifies with gauge–Higgs unification — the Higgs as the internal weak gauge component

— whose gauge-current (not metric-curl) portal *derives the bare-Casimir normalization*, the distinction that separates 0.223 from the curvature-shifted alternatives, while the Schur block itself remains the single-portal identification;

- a normal-form theorem shows that these ingredients — gauge masslessness, Casimir factorization, a single generating operator, no free constant — force $\mathcal{A}(J)$ uniquely, so the weak angle is a theorem about $SU(2)$ representation theory once the single-portal condition (hence $c = 1$) holds;
- the whole structure organizes as a probe-energy ladder $D = 11 \rightarrow 10 \rightarrow 9$ — with $D = 10$ the rung we live on, the photon protected by $Q\eta_0 = 0$, the electromagnetic coupling read at the $D = 9$ boundary, and the negative branch playing the role of the electroweak order parameter, landing, with the same single scale, near the Higgs and top masses.

The conjectural content has been reduced and located. Gauge–Higgs unification derives the operator content of the de Vries block (the single-portal Schur structure and the bare-Casimir normalization), leaving two sharp residues: the Hosotani-potential coefficient that sets the magnitude of the negative diagonal, and the identification of the W, Z poles with the Schur eigenvalues. Everything else is a theorem or a standard reduction. The weak angle is not obtained from the M^{pqr} coupling-ratio fit, whose six-digit match between a four-integer family and a three-digit datum is an over-determination of the data (Section 15); it is obtained from the representation theory of $\mathcal{A}(J)$, where there are no integers to choose — and we hold our own $c = 1$ to that same standard (Section 8.8).

The construction will stand or fall on computations that are now well-posed: the Hosotani-potential calculation on the weak $\mathbb{CP}^1/S^3$ (fixing the diagonal coefficient), the justification of the Schur-pole reading of the W, Z masses, and the search for (or exclusion of) the predicted neutral state near 96.5 GeV. If these succeed, the weak mixing angle will have joined the short list of Standard Model numbers with a geometric origin. If they fail, the construction falls on identified assumptions, not on a thicket of adjustable ones.

A The algebra of the branch operator

We collect elementary facts about $\mathcal{A}(J)$ used in the main text.

The characteristic polynomial of $\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}$ is $X^2 + JX - J$, with roots

$$X_+(J) = \frac{-J + \sqrt{J^2 + 4J}}{2} > 0, \quad X_-(J) = \frac{-J - \sqrt{J^2 + 4J}}{2} < 0, \quad (86)$$

and $X_+X_- = -J$, $X_+ + X_- = -J$. As $J \rightarrow 0$, $X_+(J) \rightarrow \sqrt{J} \rightarrow 0$ and $X_-(J) \rightarrow -\sqrt{J} \rightarrow 0$, so the $j = 0$ slot is doubly massless; as $J \rightarrow \infty$, $X_+(J) \rightarrow 1^-$ and $X_-(J) \rightarrow -(J + 1)$, so the positive branch saturates while the negative branch grows linearly.

Resolvent/seesaw form. The positive root satisfies the fixed-point equation

$$X = \frac{J}{J + X}, \quad (87)$$

exhibiting $\mathcal{A}(J)$ as the Schur complement of a two-channel operator in which a heavy normal channel of gap J is integrated out, leaving an effective inverse propagator $\Gamma(X) = X - J/(J + X)$ for the light gauge channel. This is the boundary-resolvent reading: the de Vries quadratic is the characteristic equation of a single integrated-out channel.

Relativistic-orbit form, and the meaning of the bound. The same positive root has a kinematic closed form that requires no negative eigenvalue at all. The secular equation $X^2 + JX - J = 0$ is *identical* to

$$\frac{\beta^2}{\sqrt{1-\beta^2}} = \sqrt{J}, \quad X_+(J) = \beta^2, \quad (88)$$

as one checks by squaring: $\beta^4 = J(1 - \beta^2) \Leftrightarrow \beta^4 + J\beta^2 - J = 0$, the de Vries quadratic in β^2 . Thus the gauge mass is $M_V = \mu\beta_V$ with β_V the relativistic orbital velocity fixed by (88): $\beta_W = 0.754$, $\beta_Z = 0.856$, giving $M_W = \mu\beta_W = 80.4$ and $M_Z = \mu\beta_Z = 91.2$ GeV. Two things are gained over the matrix reading. First, $X_+ = \beta^2$ is manifestly in $(0, 1)$, so the bound $X_+(J) < 1$ that keeps the electroweak masses below the compactification scale is nothing but the relativistic bound $\beta < 1$; the boundedness has a physical cause. Second, the spectrum is positive-definite — there is no tachyonic partner and no ghost — so the negative eigenvalue of the matrix form is seen to be an artifact of writing a manifestly bounded, positive quantity as one root of a quadratic. The dynamical content to be supplied is then the quantization rule (88), $\beta^2\gamma = \sqrt{C_2}$, relating the internal orbital velocity to the SU(2) Casimir — a de Broglie / Bohr–Sommerfeld condition for a mode circulating the internal SU(2).

Deformed coefficient. For $\mathcal{A}_c(J)$ of (19) the positive root is $X_+(J; c) = \frac{1}{2}(-cJ + \sqrt{c^2J^2 + 4J})$ and

$$\sin^2 \theta_W(c) = 1 - \frac{X_+(3/4; c)}{X_+(2; c)}, \quad (89)$$

tabulated in Table 1; the de Vries point $c = 1$ has $d \sin^2 \theta_W / dc = -0.140$.

Weak-angle and α identities. At $c = 1$,

$$X_+(3/4) = \frac{1}{8}(-3 + \sqrt{57}) = 0.568729 \dots, \quad (90)$$

$$X_+(2) = -1 + \sqrt{3} = 0.732051 \dots, \quad (91)$$

$$\sin^2 \theta_W = 1 - \frac{X_+(3/4)}{X_+(2)} = 0.223101 \dots, \quad (92)$$

$$\frac{1}{\alpha} = \frac{4\pi}{X_+(3/4)[X_+(2) - X_+(3/4)]} = 135.288 \dots \quad (93)$$

Note $|X_-(2)| = 1 + \sqrt{3} = 2.732 \dots$, so the negative $j = 1$ root is $-(1 + \sqrt{3})$, an exact surd; this underlies the vacuum-scale identity $v_{EW} = \sqrt{2}\mu\sqrt{1 + \sqrt{3}}$ used in Section 11, giving $v_{EW} = 249$ GeV for $\mu = 106.6$ GeV.

B Harmonic analysis on $S^3 \cong \text{SU}(2)$

We record the spectral facts used in Sections 4–5.

Scalars. On the unit round S^3 , $\Delta_0 Y_\ell = \ell(\ell + 2)Y_\ell$, $\ell = 0, 1, 2, \dots$, multiplicity $(\ell + 1)^2$. With $\ell = 2j$ this is $\Delta_0 = 4j(j + 1) = 4J$ on the Peter–Weyl block $V_j \otimes V_j^*$, dimension $(2j + 1)^2$, $j \in \{0, \frac{1}{2}, 1, \dots\}$.

Co-exact one-forms. The transverse vector modes are co-exact one-forms, on which the curl $C = \star d$ is self-adjoint with $C^2 = \Delta_1$. The eigenvalues are $C\omega = \pm(n + 1)\omega$, $n = 1, 2, \dots$, so $\Delta_1 = (n + 1)^2$; the signed (chiral) spectrum is the defining feature that distinguishes C from $\sqrt{\Delta_1}$ and supplies the first-order off-diagonal.

Casimir factorization on SU(2). Writing the left-invariant coframe e_a and generators $T_a^{(j)}$, the operator $A_j = \sum_a T_a^{(j)} \otimes e_a$ obeys $A_j^\dagger A_j = \sum_a T_a T_a = J$ in the orthonormal coframe, which is the abstract content of Lemma 3.1 and the reason the off-diagonal carries unit coefficient.

Why SU(2) and not SO(3). $L^2(\text{SO}(3)) = \bigoplus_{j \in \mathbb{Z}_{\geq 0}} V_j \otimes V_j^*$ contains *only integer* j ; the half-integer $j = \frac{1}{2}$ (W slot) appears in $L^2(\text{SU}(2))$ but not in $L^2(\text{SO}(3))$. The W at $j = \frac{1}{2}$ is thus a direct probe of the global structure of the internal three-manifold.

C The Freund–Rubin linearization, schematically

We outline the structure of (53). On $\text{AdS}_4 \times X_7$ with $F_{\mu\nu\rho\sigma} = 3m \epsilon_{\mu\nu\rho\sigma}$, the linearized equations for internal fluctuations couple the metric, three-form, and their internal form-components. The flux enters the three-form fluctuation equation through a term $\sim m \star_7 d_7$ acting on internal p -forms, which raises/lowers form-degree by one and is first-order in internal derivatives. For the pair (internal one-form a , scalar partner ϕ) the linearized mass operator is, schematically,

$$\mathcal{M} = \begin{pmatrix} \Delta_0 & \alpha m \star d \\ \alpha m (\star d)^\dagger & \Delta_1 - \beta m^2 \end{pmatrix}, \quad (94)$$

the off-diagonal flux mixing and the $-\beta m^2$ flux/curvature shift being the two structures highlighted in the classic spectra [9, 8, 10]. On the spin- j sector of the unit electroweak S^3 , $\Delta_0 = 4J$, $\star d \sim \sqrt{\Delta_1} \sim \sqrt{J}$ on the matched mode, and the Einstein relation between m and the S^3 curvature renders the shift proportional to J . After projecting onto the (gauge, order-parameter) two-channel subspace, restoring gauge masslessness of the transverse vector, and rescaling by the overall μ^2 , the block becomes $\mathcal{A}_c(J)$ of (19) with c the dimensionless ratio (56). We do not claim to have evaluated α, β and shown $c = 1$; that is the open computation (21). The appendix is included to make the *structure* — not the coefficient — explicit and checkable.

D The weak-angle scan and over-determination

To let the reader inspect the over-determination critique of Section 15, we record the M^{pqr} weak-angle map. The shape parameter $\beta(q/p)$ solves the Bailin–Love cubic (83); the ratio is $g_Y^2/g_2^2 = \frac{3}{2}(p^2 n^2/r^2)F(\beta)$ with $F(\beta) = (3-4\beta)^2(1+\beta)/(1-2\beta)$. Table 3 lists the selected point and two nearby integer alternatives, illustrating that small integer changes traverse the target value, i.e. the family is dense near 0.287 and the six-digit match carries no selection information.

(p, q, r, n)	β	g_Y^2/g_2^2	rel. dev. from $\tan^2 \theta_{\text{dV}}$	status
(12, 14, 91, 1)	0.32848	0.287170	2.7×10^{-6}	“selected” (a fit)
nearby (p, q, r, n)	—	spans 0.287	arbitrarily small	shows density

Table 3: The M^{pqr} weak-angle scan. A four-integer family reproduces a three-digit datum to six digits; the precision of the match reflects the density of the family, not a prediction. Cf. Section 15.

The contrast with the parameter-free route is the whole point: in $\mathcal{A}(J)$ there are no integers to scan. The inputs are the spins $j = \frac{1}{2}, 1$ (fixed by the electroweak content) and the flux normalization $c = 1$ (a single falsifiable statement), and the output (5) is forced.

E Numerical summary

Quantity	Value (this model)	Comparison
$X_+(3/4)$	0.5687293	—
$X_+(2)$	0.7320508	$= \sqrt{3} - 1$
$\sin^2 \theta_W = 1 - X_+(3/4)/X_+(2)$	0.2231013	exp. 0.22305(20)
$\tan^2 \theta_W$	0.2871691	—
$1/\alpha$ (Eq. (10))	135.288	$1/\alpha(0) = 137.036$
μ (from M_Z)	106.58 GeV	$\simeq \sqrt{v M_Z}/2$
$M_W = \mu \sqrt{X_+(3/4)}$	80.374 GeV	exp. 80.377(12)
$\mu \sqrt{ X_-(3/4) }$	122.4 GeV	$m_H = 125.25$
$\mu \sqrt{ X_-(2) }$	176.2 GeV	$m_t = 172.7$
$M_{3/2} = \mu \sqrt{X_+(15/4)}$	96.5 GeV	prediction (F2)
$v_{EW} = \sqrt{2} \mu \sqrt{1 + \sqrt{3}}$	249 GeV	exp. 246.2

F The complex-pole sharpening

We record how the small residual between the predicted $\sin^2 \theta_W = 0.22310$ and the measured on-shell 0.22305(20) is consistent with the pole reading of Section 2.5. The physical gauge-boson masses are defined through the complex pole position $s_V = M_V^2 - iM_V \Gamma_V$ of the propagator, and two on-shell conventions circulate: the “running” mass $\overline{M}_V$ (the real part of $\sqrt{s_V}$ in one scheme) and the experimentalists’ on-shell mass M_V used in the Breit–Wigner fit. The two differ by

$$M_V - \overline{M}_V \simeq \frac{\Gamma_V^2}{2M_V}, \quad (95)$$

which for the Z ($\Gamma_Z = 2.495$ GeV) is about 34 MeV and for the W ($\Gamma_W = 2.085$ GeV) about 27 MeV. Translating into the ratio,

$$\delta \sin^2 \theta_W \sim \frac{2M_W^2}{M_Z^2} \left(\frac{\delta M_W}{M_W} - \frac{\delta M_Z}{M_Z} \right) \sim \text{few} \times 10^{-4}, \quad (96)$$

which is the size of the residual $0.22310 - 0.22305 = 5 \times 10^{-5}$. The de Vries number 0.22310 should therefore be compared to the on-shell value in the *pole* convention, and at that level the agreement is within the convention ambiguity. We stress that this is a consistency statement, not a postdiction of the residual; the point is only that a 5×10^{-5} residual is below the scheme ambiguity and does not constitute a tension.

G The deformation family and the two-spin tower

For completeness we record the two generalizations of the relation used in the text. The coefficient deformation (19) gives $\sin^2 \theta_W(c)$ in Table 1; the special value $c = 1$ is the self-dual point, and $c = \frac{1}{2}$ is the conformal-weight value that fails. The two-spin tower replaces the electroweak pair $(\frac{1}{2}, 1)$ by a general pair (j_a, j_b) ,

$$\sin^2 \theta(j_a, j_b) = 1 - \frac{X_+(J_a)}{X_+(J_b)}, \quad J_i = j_i(j_i + 1), \quad (97)$$

tabulated in Table 5. The de Vries point is $(j_a, j_b) = (\frac{1}{2}, 1)$. The monotonicity of X_+ guarantees $0 < \sin^2 \theta(j_a, j_b) < 1$ for $j_a < j_b$, and the large-spin behaviour $X_+(J) \rightarrow 1$ makes the family accumulate near $\sin^2 \theta \rightarrow 0$ as both spins grow, preserving the ordering of the lowest rungs.

(j_a, j_b)	J_a, J_b	$\sin^2 \theta(j_a, j_b)$
$(\frac{1}{2}, 1)$	$\frac{3}{4}, 2$	0.22310 (electroweak)
$(\frac{1}{2}, \frac{3}{2})$	$\frac{3}{4}, \frac{15}{4}$	0.30684
$(1, \frac{3}{2})$	$2, \frac{15}{4}$	0.10778
$(\frac{1}{2}, 2)$	$\frac{3}{4}, 6$	0.34852

Table 5: The two-spin tower $\sin^2 \theta(j_a, j_b) = 1 - X_+(J_a)/X_+(J_b)$. The electroweak case is the first row. The remaining entries are the predictions of the same operator for any extended sector that splits along higher internal spins.

H A status glossary

To keep the epistemic status of each claim legible, we used four tags in the text. We collect their meanings here.

- *Theorem*: a mathematical statement proved in this paper (Theorem 3.2, Lemma 3.1, Corollary 3.3).
- *Derived*: a consequence of established physics applied to the stated setting (the Weinberg coupling rule, the photon masslessness (73), the global-form descent of Section 13, the curl spectrum of Section 5.3).
- *Structural*: forced once a stated assumption is granted (the reduction of the flux block to $\mathcal{A}_c(J)$, Section 8).
- *Conjectural*: a physical hypothesis not yet established, of which there is exactly one load-bearing instance — the flux normalization $c = 1$, Eq. (21) — together with the optional scenarios (Higgs as radion, the negative-branch identification of m_H, m_t) that are attractive but unproven.

I On the use of a large language model in preparing this paper

This paper was prepared with the assistance of a large language model (Claude). In the interest of transparency we disclose this plainly, and we use this appendix to record the methodological standard that governed the work — the criterion by which a geometric account of the weak mixing angle should be judged to succeed or fail — since that standard is, in effect, a statement about how the model’s output was disciplined against the characteristic failure mode of this subject (plausible-sounding numerology).

What the model did and did not do. The model assembled and organized the argument, performed the elementary algebra and the numerical evaluations (all of which are reproducible from the formulae in the text), drafted the prose, and — importantly — diagnosed and excised a specific drift in the prior research record, namely the temptation to “derive” the weak angle from a four-integer M^{pqr} fit (Section 15). The model did *not* establish the one load-bearing physical conjecture — it did not show that the order parameter satisfies the single-portal hypothesis — but it did sharpen $c = 1$ from a bare ansatz to the conditional derivation of Proposition 8.3, including the result that the Weitzenböck curvature shift cancels in the ratio under the single-portal condition and otherwise yields the explicit j -dependent failure $c = (4J + 1)/(4J)$ (angle 0.267). That conditional derivation was obtained and then *independently cross-checked by two different language models* (the present one and a second, GPT-class model), which reached the same conclusion and the same numbers ($c = \beta/\alpha^2$; $c = 1$ iff $\beta = \alpha^2$; failure value $\frac{4}{3}, \frac{9}{8}$); the literature checks were performed by online search rather than from training memory. The

gauge–Higgs identification of Section 8.7 was checked the same way, and here the two models *disagreed* in a productive manner: one judged that gauge–Higgs unification delivers the full Schur block, while the other, performing the explicit S^2 mode reduction, found that the standard gauge–Higgs spectrum is a Goldstone plus a separate scalar and that the Schur block is an additional identification. We adopted the more skeptical, explicitly computed conclusion; the disagreement, and its resolution by direct calculation, is the reason the section claims only the bare-Casimir normalization for gauge–Higgs unification and lists the Schur identification among the residues. No claim in this paper should be accorded authority on the basis of its having been produced with model assistance; each claim stands or falls on the tag it carries (Appendix H) and on the elementary mathematics displayed.

The criterion of success. It is worth stating, before any computation, what would count as success and what would count as failure, because the subject is one in which the absence of a sharp criterion has historically allowed wishful thinking to masquerade as progress — a hazard to which an articulate text generator is at least as susceptible as a human author.

A successful geometric account of the weak mixing angle must satisfy three conditions. First, it must produce the number $\sin^2 \theta_W = 0.2231$ *without adjusting a continuous parameter to it*; a construction with a dial that is turned until the answer comes out is a re-description, not an explanation. Second, it must produce it from data that are independently motivated: the internal symmetry must be the electroweak $SU(2)$ for an independent reason, the representations must be the physical ones, and the single dynamical assumption (if any) must be stateable in one line and testable. Third, the same structure must be *rigid* — small deformations must move the prediction, so that the agreement is a genuine constraint and not a soft coincidence compatible with a range of theories. We built the rigidity test into the construction from the start: the deformation (19) and its sensitivity (20) are precisely the statement that the prediction is sharp.

The corresponding failure modes are equally important to name. The construction fails if (a) the weak angle can only be obtained by fitting discrete data (the over-determination trap of Section 15), (b) the half-integer label $j = \frac{1}{2}$ can only be installed by an ad hoc change of field content, or (c) the negative diagonal $-J$ requires an unmotivated tachyon rather than a flux mechanism present in the parent theory. The $S^3 \cong SU(2)$ geometry passes (b) by Peter–Weyl and (c) by Freund–Rubin flux, and the parameter-free route passes (a) by construction.

Finally, we are clear about what is *not* part of the success criterion, because over-ambition is its own failure. We do not require the construction to predict the absolute electroweak scale, the fermion masses, or the cosmological constant. The weak mixing angle is a single, sharp, dimensionless number; explaining it geometrically, with no tuned parameter, would already be a result worth having, in the same way that explaining the quantization of electric charge by a compact $U(1)$ is a result even though it says nothing about the value of e . We aim at the angle and at the mass ratio, and we are explicit about the rest.

On over-determination and the avoidance of integer fits. The over-determination demonstrated quantitatively in Section 15.3 is an instance of a recurring failure mode that this construction is designed to avoid. A program that begins with the clean, parameter-free relation (5) can be drawn into “deriving” it from a multi-integer geometric family that reproduces it to many digits; the many digits then become load-bearing, and the construction is built to protect a coincidence of the *fit* rather than of the physics. The tell-tale signs are (i) the headline number’s quoted precision far exceeds the precision of the data it is matched to, (ii) the selecting integers have no independent derivation, and (iii) the rest of the construction accretes undetermined coefficients — shape moduli, flux periods, potential constants — that are never computed, while the integer fit is treated as established. The discipline adopted here is to obtain the weak angle from the parameter-free $\mathcal{A}(J)$, to use M^{pqr} only for its isometries, topology, and charge lattice, and to isolate the single dynamical assumption $c = 1$ so that it

can be tested rather than buried.

An observation on the literature. The chirality argument of Section 14 — that the obstruction to chiral fermions selects $D = 10$ as the physical rung, with $D = 11$ chirality-impossible (smooth no-go) and $D = 9$ chirality-trivial (vector-like residual group) — is assembled entirely from textbook results, yet we have not found it stated as an organizing principle in the standard Kaluza–Klein or string-phenomenology literature, where the individual facts appear in isolation. We record this only as an observation: the synthesis is elementary, and its apparent absence from the standard presentations is a gap in exposition rather than evidence of anything more; we make no claim beyond that the framing seems not to be commonly emphasized. We mention it here, in the methodological appendix, rather than in the main text, because it is a remark about the literature and not a result.

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